

ground pivots at each time step create a time-varying shaking couple that rocks the ground plane. These pin forces can be larger due to the balance weights and if so will increase the shaking couple compared to its former value in the unbalanced linkage—one trade-off for reducing the shaking forces to zero. The stresses in the links and pins may also increase as a result of force-balancing.

12.5 EFFECT OF BALANCING ON INPUT TORQUE

Individually balancing a link which is in pure rotation by the addition of a counterweight will have the side effect of increasing its mass moment of inertia. The "flywheel effect" of the link is increased by this increase in its moment of inertia. Thus the torque needed to accelerate that link will be greater. The input torque will be unaffected by any change in the I of the input crank when it is run at constant angular velocity. But, any rockers in the mechanism will have angular accelerations even when the crank does not. Thus, individually balancing the rockers will tend to increase the required input torque even at constant input crank velocity.

Adding counterweights to the rotating links, necessary to force balance the entire linkage, both increases the links mass moment of inertia and also (individually) *unbalances* those rotating links in order to gain the global balance. Then the CGs of the rotating links will not be at their fixed pivots. Any angular acceleration of these links will add to the torque loading on the linkage. Balancing an entire linkage by this method then can have the side effect of increasing the variation in the required input torque. A larger flywheel may be needed on a balanced linkage in order to achieve the same coefficient of fluctuation as the unbalanced version of the linkage.

Figure 12-8 shows the input torque curve for the unbalanced linkage and for the same linkage after complete force-balancing has been done. The peak value of the required input torque has increased as a result of force-balancing.

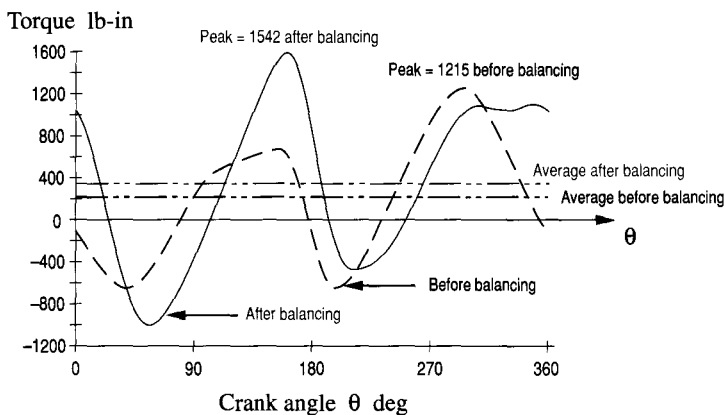
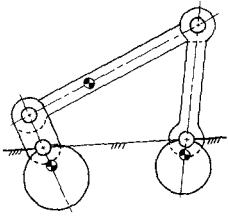


FIGURE 12-8

Unbalanced and balanced input torque curves for the fourbar linkage of Figure 12-5

**FIGURE 12-9**

An inline fourbar linkage (6) with optimally located circular counterweights. (5)

12.6 BALANCING THE SHAKING MOMENT IN LINKAGES

The shaking moment M_s in a force-balanced linkage is the sum of the reaction torque T_{21} and the shaking couple (ignoring any externally applied loads).^{[6]*}

$$M_s = T_{21} + (R_1 \times F_{41}) \quad (12.9)$$

where T_{21} is the negative of the driving torque T_{12} , R_1 is the position vector from O_2 to O_1 (i.e., link 1), and F_{41} is the force of the rocker on the ground plane. In a general linkage, the magnitude of the shaking moment can be reduced but cannot be eliminated by means of mass redistribution within its links. Complete balancing of the shaking moment requires the addition of supplementary links and then only works for certain special configurations of a fourbar linkage.^[7]

Many techniques have been developed that use optimization methods to find a linkage mass configuration that will minimize the shaking moment alone or in combination with minimizing shaking force and/or input torque. Hockey [8], [9] shows that the fluctuation in kinetic energy and input torque of a mechanism may be reduced by proper distribution of mass within its links and that this approach is more weight efficient than adding a flywheel to the input shaft. Berkof [10] also describes a method to minimize the input torque by internal mass rearrangement. Lee and Cheng [11] and Qi and Pennestri [12] show methods to optimally balance the combined shaking force, shaking moment, and input torque in high-speed linkages by mass redistribution and addition of counterweights. Porter [13] suggests using a genetic algorithm to optimize the same set of parameters. Bagci [14] describes several approaches to balancing shaking forces and shaking moments in the fourbar slider-crank linkage. Most of these methods require significant computing resources, and space does not permit a complete discussion of them all here. The reader is directed to the references for more information.

Berkof's method for complete moment balancing of the fourbar linkage [7] is simple and useful even though it is limited to "inline" linkages, i.e., those whose link CGs lie on their respective link centerlines as shown in Figure 12-9. This is not an overly restrictive constraint since many practical linkages are made with straight links. Even if a link must have a shape that deviates from its line of centers, its CG can still be placed on that line by adding mass to the link in the proper location, increased weight being the trade-off.

For complete moment balancing, in addition to being an inline linkage, the coupler must be reconfigured to become a physical pendulum such that it is dynamically equivalent to a lumped mass model as shown in Figure 12-10. The coupler is shown in Figure

* Note that this statement is only true if the linkage is force balanced which makes the moment of the shaking couple a free vector. Otherwise it is referenced to the chosen global coordinate system. See reference [6] for complete derivations of the shaking moment for both force-balanced and unbalanced linkages.

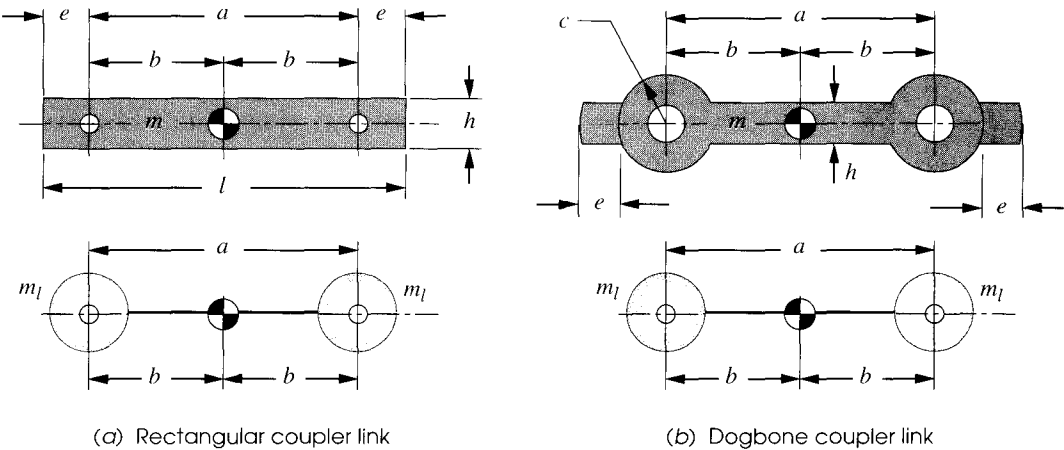


FIGURE 12-10

Making the coupler link a physical pendulum

12-10a as a uniform rectangular bar of mass m , length a and width h and in Figure 12-10b as a “dogbone.” These are only two of many possibilities. We want the lumped masses to be at the pivot pins, connected by a “massless” rod. Then the coupler’s lumped masses will be in pure rotation either as part of the crank or as part of the rocker.* This can be accomplished by adding mass as indicated by dimension e at the coupler ends.

The three requirements for dynamic equivalence were stated in Section 10.2 (p. 492) and are: equal mass, same CG location, and same mass moment of inertia. The first and second of these are easily satisfied by placing $m_l = m/2$ at each pin. The third requirement can be stated in terms of radius of gyration k instead of moment of inertia using equation 10.10 (p. 498).

$$k = \sqrt{\frac{I}{m}} \tag{12.10}$$

Taking each lump separately as if the massless rod were split at the CG into two rods each of length b , the moment of inertia I_l of each lump will be

and

$$I_l = \frac{I}{2} = m_l b^2$$
$$I = 2m_l b^2 = mb^2 \tag{12.11a}$$

then

$$k = \sqrt{\frac{mb^2}{m}} = b = \frac{a}{2} \tag{12.11b}$$

For the link configuration in Figure 12-10a, this will be satisfied if the link dimensions have the following dimensionless ratios.

$$\frac{e}{h} = \frac{1}{2} \sqrt{3 \left(\frac{a}{h} \right)^2 - 1} - \frac{a}{2h} \tag{12.12a}$$

* Note that this arrangement also makes each pin joint the center of percussion for the other pin as the center of rotation. This means that a force applied at either pin will have a zero reaction force at the other pin, effectively decoupling them dynamically. See Section 10.8 (p. 498) and also Figure 13-10 (p. 616) for further discussion of this effect.

$$\frac{l}{h} = \frac{a}{h} + 2\frac{e}{h} \quad (12.12b)$$

where e defines the length of the material that must be added at each end to satisfy equation 12.11b.

For the link configuration in Figure 12-10b, the length e of the added material of width h needed to make it a physical pendulum can be found from

$$A\left(\frac{e}{h}\right)^3 + B\left(\frac{e}{h}\right)^2 + C\left(\frac{e}{h}\right) + D = 0 \quad (12.13)$$

where :

$$A = 8$$

$$B = 12\left(\frac{a}{c}\right) + 24$$

$$C = 24\left(\frac{a}{c}\right) + 26$$

$$D = -2\left(\frac{a}{c}\right)^3 + 13\left(\frac{a}{c}\right) + 12\pi - 10$$

The second step is to force balance the linkage with its modified coupler using the method of Section 12.3 (p.579) and define the required counterweights on links 2 and 4. With the shaking forces eliminated, the shaking moment is a free vector, as is the input torque.

Then as the third step, the shaking moment can be counteracted by adding geared inertia counterweights to links 2 and 4 as shown in Figure 12-11. These must turn in the opposite direction to the links, so they require a gear ratio of -1 . Such an inertia coun-

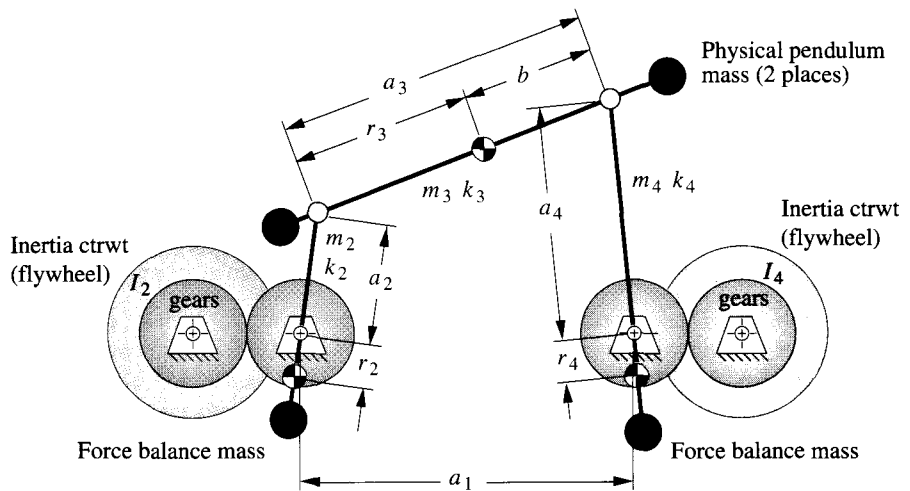


FIGURE 12-11

Completely force and moment balanced inline fourbar linkage with physical pendulum coupler and inertia counterweights on rotating links

terweight can balance any planar moment that is proportional to an angular acceleration and does not introduce any net inertia, forces to upset the force balance of the linkage. Trade-offs include increased input torque and larger pin forces resulting from the torque required to accelerate the additional rotational inertia. There can also be large loads on the gear teeth and impact when torque reversals take up the gearsets' backlash, causing noise.

The shaking moment of an inline fourbar linkage is derived in reference [6] as

$$M_s = \sum_{i=2}^4 A_i \alpha_i \quad (12.14)$$

where: $A_2 = -m_2(k_2^2 + r_2^2 + a_2 r_2)$

$$A_3 = -m_3(k_3^2 + r_3^2 - a_3 r_3)$$

$$A_4 = -m_4(k_4^2 + r_4^2 + a_4 r_4)$$

α_i is the angular acceleration of link i . The other variables are defined in Figure 12-11.

Adding the effects of the two inertia counterweights gives

$$M_s = \sum_{i=2}^4 A_i \alpha_i + I_2 \alpha_2 + I_4 \alpha_4 \quad (12.15)$$

The shaking moment can be forced to zero if

$$\begin{aligned} I_2 &= -A_2 \\ I_4 &= -A_4 \\ A_3 &= 0, \quad \text{or} \quad k_3^2 = r_3(a_3 - r_3) \end{aligned} \quad (12.16)$$

This leads to a set of five design equations that must be satisfied for complete force and moment balancing of an inline fourbar linkage.

$$m_2 r_2 = m_2 b \left(\frac{a_2}{a_3} \right) \quad (12.17a)$$

$$m_4 r_4 = m_3 r_3 \left(\frac{a_4}{a_3} \right) \quad (12.17b)$$

$$k_3^2 = r_3 b \quad (12.17c)$$

$$I_2 = m_2(k_2^2 + r_2^2 + a_2 r_2) \quad (12.17d)$$

$$I_4 = m_4(k_4^2 + r_4^2 + a_4 r_4) \quad (12.17e)$$

Equations 12.17a and 12.17b are the force-balance criteria of equation 12.8 written for the inline linkage case. Equation 12.17c defines the coupler as a physical pendulum. Equations 12.17d and 12.17e define the mass moments of inertia required for the two inertia counterweights. Note that if the linkage is run at constant angular velocity, a_2 will be zero in equation 12.14 and the inertia counterweight on link 2 can be omitted.

12.7 MEASURING AND CORRECTING IMBALANCE

While we can do a great deal to ensure balance when designing a machine, variations and tolerances in manufacturing will preclude even a well-balanced design from being in perfect balance when built. Thus there is need for a means to measure and correct the imbalance in rotating systems. Perhaps the best example assembly to discuss is that of the automobile tire and wheel, with which most readers will be familiar. Certainly the design of this device promotes balance, as it is essentially cylindrical and symmetrical. If manufactured to be perfectly uniform in geometry and homogeneous in material, it should be in perfect balance as is. But typically it is not. The wheel (or rim) is more likely to be close to balanced, as manufactured, than is the tire. The wheel is made of a homogeneous metal and has fairly uniform geometry and cross section. The tire, however, is a composite of synthetic rubber elastomer and fabric cord or metal wire. The whole is compressed in a mold and steam-cured at high temperature. The resulting material varies in density and distribution, and its geometry is often distorted in the process of removal from the mold and cooling.

STATIC BALANCING After the tire is assembled to the wheel, the assembly must be balanced to reduce vibration at high speeds. The simplest approach is to statically balance it, though it is not really an ideal candidate for this approach as it is thick axially compared to its diameter. To do so it is typically suspended in a horizontal plane on a cone through its center hole. A bubble level is attached to the wheel, and weights are placed at positions around the rim of the wheel until it sits level. These weights are then attached to the rim at those points. This is a single-plane balance and thus can only cancel the unbalanced forces. It has no effect on any unbalanced moments due to uneven distribution of mass along the axis of rotation. It also is not very accurate.

DYNAMIC BALANCING The better approach is to dynamically balance it. This requires a dynamic balancing machine be used. Figure 12-12 shows a schematic of such a device used for balancing wheels and tires or any other rotating assembly. The assembly to be balanced is mounted temporarily on an axle, called a mandrel, which is supported in bearings within the balancer. These two bearings are each mounted on a suspension which contains a transducer that measures dynamic force. A common type of force transducer contains a piezoelectric crystal which delivers a voltage proportional to the force applied. This voltage is amplified electronically and delivered to circuitry or software which can compute its peak magnitude and the phase angle of that peak with respect to some time reference signal. The reference signal is supplied by a shaft encoder on the mandrel which provides a short duration electrical pulse once per revolution in exactly the same angular location. This encoder pulse triggers the computer to begin processing the force signal. The encoder may also provide some large number of additional pulses equispaced around the shaft circumference (often 1024). These are used to trigger the recording of each data sample from the transducers in exactly the same location around the shaft and to provide a measure of shaft velocity via an electronic counter.

The assembly to be balanced is then "spun up" to some angular velocity, usually with a friction drive contacting its circumference. The drive torque is then removed and the drive motor stopped, allowing the assembly to "freewheel." (This is to avoid measuring any forces due to imbalances in the drive system.) The measuring sequence is begun, and the dynamic forces at each bearing are measured simultaneously and their waveforms stored. Many cycles can be measured and averaged to improve the quality

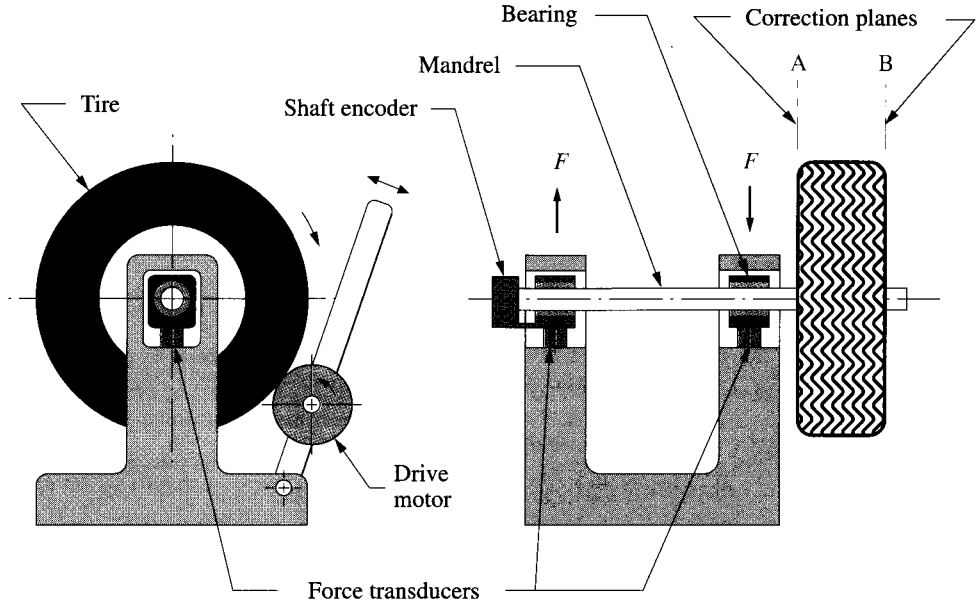


FIGURE 12-12

A dynamic wheel balancer

of the measurement. Because forces are being measured at two locations displaced along the axis, both summation of moment and summation of force data are computed.

The force signals are sent to a built-in computer for processing and computation of the needed balance masses and locations. The data needed from the measurements are the magnitudes of the peak forces and the angular locations of those peaks with respect to the shaft encoder's reference angle (which corresponds to a known point on the wheel). The axial locations of the wheel rim's inside and outside edges (the correction planes) with respect to the balance machine's transducer locations are provided to the machine's computer by operator measurement. From these data the net unbalanced force and net unbalanced moment can be calculated since the distance between the measured bearing forces is known. The mass-radius products needed in the correction planes on each side of the wheel can then be calculated from equations 12.3 (p. 574) in terms of the mR product of the balance weights. The correction radius is that of the wheel rim. The balance masses and angular locations are calculated for each correction plane to put the system in dynamic balance. Weights having the needed mass are clipped onto the inside and outside wheel rims (which are the correction planes in this case), at the proper angular locations. The result is a fairly accurately dynamically balanced tire and wheel.

12.8 REFERENCES

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12.9 PROBLEMS

- *†12-1 A system of two coplanar arms on a common shaft, as shown in Figure 12-1 (p. 573), is to be designed. For the row(s) assigned in Table P12-1, find the shaking force of the linkage when run unbalanced and design a counterweight to statically balance the system. Work in any consistent units system you prefer.
- †12-2 The minute hand on Big Ben weighs 40 lb and is 10 ft long. Its CG is 4 ft from the pivot. Calculate the mR product and angular location needed to statically balance this link and design a physical counterweight, positioned close to the center. Select material and design the detailed shape of the counterweight which is of 2 in uniform thickness in the Z direction.
- †12-3 A "V for victory" advertising sign is being designed to be oscillated about the apex of the V, on a billboard, as the rocker of a fourbar linkage. The angle between the legs of the V is 20°. Each leg is 8 ft long and 1.5 ft wide. Material is 0.25 in-thick aluminum. Design the V link for static balance.
- †12-4 A three-bladed ceiling fan has 1.5 ft by 0.25 ft equispaced rectangular blades that nominally weigh 2 lb each. Manufacturing tolerances will cause the blade weight to vary up to plus or minus 5%. The mounting accuracy of the blades will vary the location of the CG versus the spin axis by plus or minus 10% of the blades' diame-

* Answers in Appendix F.

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs.

TABLE P12-1 Data for Problem 12-1

Row	m_1	m_2	R_1	R_2
a.	0.20	0.40	1.25 @ 30°	2.25 @ 120°
b.	2.00	4.36	3.00 @ 45°	9.00 @ 320°
c.	3.50	2.64	2.65 @ 100°	5.20 @ -60°
d.	5.20	8.60	7.25 @ 150°	6.25 @ 220°
e.	0.96	3.25	5.50 @ -30°	3.55 @ 120°

ters. Calculate the weight of the largest steel counterweight needed at a 2-in radius to statically balance the worst-case blade assembly.

*†12-5 A system of three noncoplanar weights is arranged on a shaft generally as shown in Figure 12-3 (p. 577). For the dimensions from the row(s) assigned in Table P12-2, find the shaking forces and shaking moment when run unbalanced at 100 rpm and specify the mR product and angle of the counterweights in correction planes A and B needed to dynamically balance the system. The correction planes are 20 units apart. Work in any consistent units system you prefer.

*†12-6 A wheel and tire assembly has been run at 100 rpm on a dynamic balancing machine as shown in Figure 12-12 (p. 591). The force measured at the left bearing had a peak of 5 lb at a phase angle of 45° with respect to the zero reference angle on the tire. The force measured at the right bearing had a peak of 2 lb at a phase angle of -120° with respect to the reference zero on the tire. The center distance between the two bearings on the machine is 10 in. The left edge of the wheel rim is 4 in from the centerline of the closest bearing. The wheel is 7 in wide at the rim. Calculate the size and location with respect to the tire's zero reference angle, of balance weights needed on each side of the rim to dynamically balance the tire assembly. The wheel rim diameter is 15 in.

*†12-7 Repeat Problem 12-6 for measured forces of 6 lb at a phase angle of -60° with respect to the reference zero on the tire, measured at the left bearing, and 4 lb at a phase angle of 150° with respect to the reference zero on the tire, measured at the right bearing. The wheel diameter is 16 in.

TABLE P12-2 Data for Problem 12-5

Row	m_1	m_2	m_3	l_1	l_2	l_3	R_1	R_2	R_3
a.	0.20	0.40	1.24	2	8	17	1.25 @ 30°	2.25 @ 120°	5.50 @ -30°
b.	2.00	4.36	3.56	5	7	16	3.00 @ 45°	9.00 @ 320°	6.25 @ 220°
c.	3.50	2.64	8.75	4	9	11	2.65 @ 100°	5.20 @ -60°	1.25 @ 30°
d.	5.20	8.60	4.77	7	12	16	7.25 @ 150°	6.25 @ 220°	9.00 @ 320°
e.	0.96	3.25	0.92	1	3	18	5.50 @ 30°	3.55 @ 120°	2.65 @ 100°

* Answers in Appendix F.

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs.

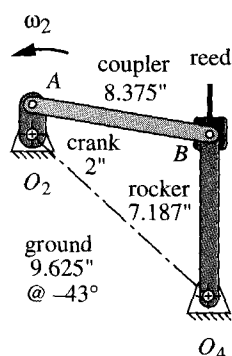


FIGURE P12-1

Problem 12-9

* Answers in Appendix F.

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKsolver* equation solver programs.

‡ These problems are suited to solution using program *FOURBAR* which is on the attached CD-ROM.

*†12-8 Table P11-3 (p. 560) shows the geometry and kinematic data for several fourbar linkages.

- For the row(s) from Table P11-3 assigned in this problem, calculate the size and angular locations of the counterbalance mass-radius products needed on links 2 and 4 to completely force balance the linkage by the method of Berkof and Lowen. Check your manual calculation with program *FOURBAR*.
- Calculate the input torque for the linkage both with and without the added balance weights and compare the results. Use program *FOURBAR*.

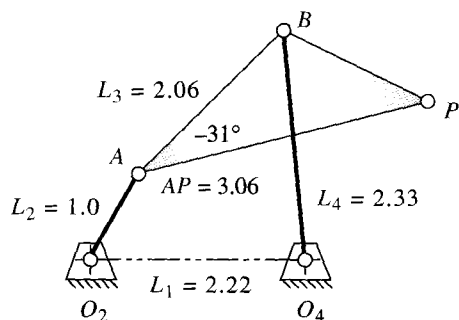
*†12-9 Link 2 in Figure P12-1 rotates at 500 rpm. The links are steel with cross sections of 1 x 2 in. Half of the 29-lb weight of the laybar and reed are supported by the linkage at point B. Design counterweights to force balance the linkage and determine its change in peak torque versus the unbalanced condition. See Problem 11-13 (p. 563) for more information on the overall mechanism.

†12-10 Figure P12-2a shows a fourbar linkage and its dimensions in meters. The steel crank and rocker have uniform cross sections of 50 mm wide by 25 mm thick. The aluminum coupler is 25 mm thick. The crank O_2A rotates at a constant speed of $\omega = 40$ rad/sec. Design counterweights to force balance the linkage and determine its change in peak torque versus the unbalanced condition.

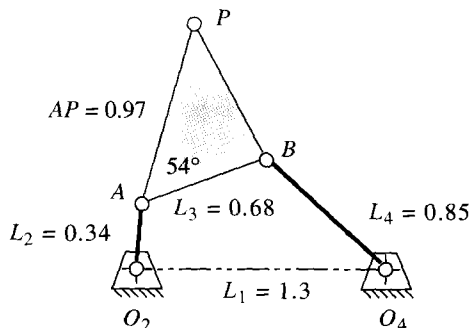
†12-11 Figure P12-2b shows a fourbar linkage and its dimensions in meters. The steel crank and rocker have uniform cross sections of 50 mm wide by 25 mm thick. The aluminum coupler is 25 mm thick. The crank O_2A rotates at a constant speed of $\omega = 50$ rad/sec. Design counterweights to force balance the linkage and determine its change in peak torque versus the unbalanced condition.

†12-12 Write a computer program or use an equation solver such as *Mathcad*, *Matlab*, or *TKsolver* to solve for the mass-radius products that will force balance any fourbar linkage for which the geometry and mass properties are known.

†12-13 Figure P12-3 shows a system with two weights on a rotating shaft. $W_1 = 15$ lb @ 0° at a 6-in radius and $W_2 = 20$ lb @ 270° at a 5-in radius. Determine the magnitudes and angles of the balance weights needed to dynamically balance the system. The balance weight in plane 3 is placed at a radius of 5 in and in plane 4 of 8 in.



(a)



(b)

FIGURE P12-2

Problems 12-10 to 12-11

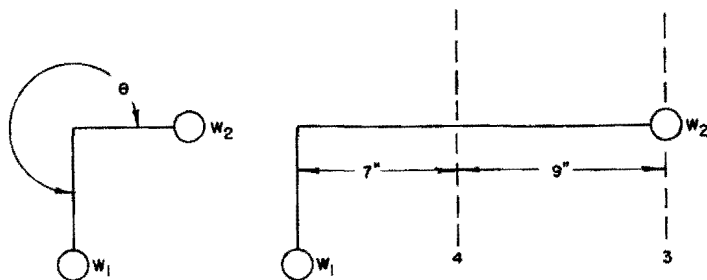


FIGURE P12-3

Problem 12-13

- †12-14 Figure P12-4 shows a system with two weights on a rotating shaft. $W_1 = 15 \text{ lb @ } 30^\circ$ at a 4-in radius and $W_2 = 20 \text{ lb @ } 270^\circ$ at a 6-in radius. Determine the radii and angles of the balance weights needed to dynamically balance the system. The balance weight in plane 3 weighs 15 lb and in plane 4 weighs 30 lb.
- †12-15 Figure P12-5 shows a system with two weights on a rotating shaft. $W_1 = 10 \text{ lb @ } 90^\circ$ at a 3-in radius and $W_2 = 15 \text{ lb @ } 240^\circ$ at a 3-in radius. Determine the magnitudes and angles of the balance weights needed to dynamically balance the system. The balance weights in planes 3 and 4 are placed at a 3-in radius.
- †12-16 Figure P12-6 shows a system with three weights on a rotating shaft. $W_1 = 9 \text{ lb @ } 90^\circ$ at a 4-in radius, $W_2 = 9 \text{ lb @ } 225^\circ$ at a 6-in radius, and $W_3 = 6 \text{ lb @ } 315^\circ$ at a 10-in radius. Determine the magnitudes and angles of the balance weights needed to dynamically balance the system. The balance weights in planes 4 and 5 are placed at a 3-in radius.
- †12-17 Figure P12-7 shows a system with three weights on a rotating shaft. $W_1 = 10 \text{ lb @ } 90^\circ$ at a 3-in radius, $W_3 = 10 \text{ lb @ } 180^\circ$ at a 4-in radius, and $W_3 = 8 \text{ lb @ } 315^\circ$ at a 4-in radius. Determine the magnitudes and angles of the balance weights needed to dynamically balance the system. The balance weight in plane 1 is placed at a radius of 4 in and in plane 5 of 3 in.

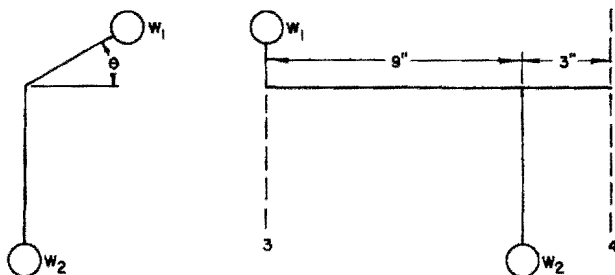
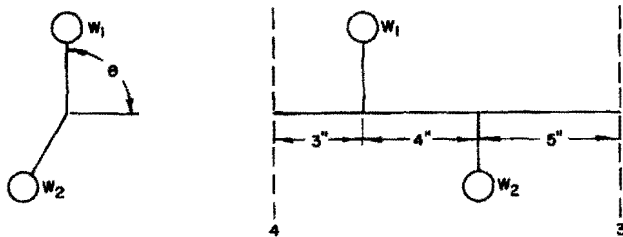


FIGURE P12-4

Problem 12-14

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs.

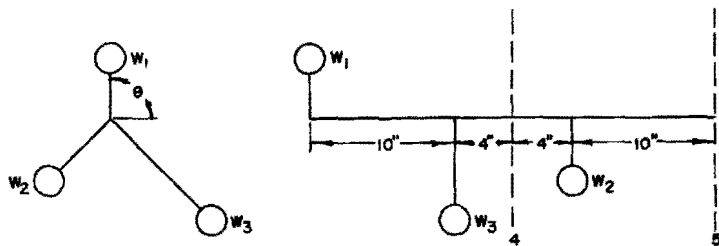
**FIGURE P12-5**

Problem 12-15

- †12-18 The drum in Figure P12-8 has been tested on a dynamic balancing machine and shows an unbalanced moment of $20,000 \text{ N}\cdot\text{mm}^2$ @ 45° about plane 1 and $15,000 \text{ N}\cdot\text{mm}^2$ @ 210° about plane 4. Determine what balance weights need to be added in planes 2 and 3 at a 150-mm radius to dynamically balance the system.
- †12-19 The drum in Figure P12-8 has been tested on a dynamic balancing machine and shows an unbalanced moment of $10,000 \text{ N}\cdot\text{mm}^2$ @ 30° about plane 1 and $25,000 \text{ N}\cdot\text{mm}^2$ @ 135° about plane 4. Determine what balance weights need to be added in planes 2 and 3 at a 100-mm radius to dynamically balance the system.
- †‡12-20 The linkage in Figure P12-9a has rectangular steel links of $20 \times 10 \text{ mm}$ cross section similar to that shown in Figure 12-10a (p. 587). Design the necessary balance weights and other features necessary to completely eliminate the shaking force and shaking moment. State all assumptions.
- †‡12-21 Repeat Problem 12-20 using links configured as in Figure 12-10b with the same cross section but having “dogbone” end diameters of 50 mm.
- †‡12-22 The linkage in Figure P12-9b has rectangular steel links of $20 \times 10 \text{ mm}$ cross section similar to that shown in Figure 12-10a (p. 587). Design the necessary balance weights and other features necessary to completely eliminate the shaking force and shaking moment. State all assumptions.
- †‡12-23 Repeat Problem 12-22 using steel links configured as in Figure 12-10b with a $20 \times 10 \text{ mm}$ cross section and having “dogbone” end diameters of 50 mm.

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKsolver* equation solver programs.

‡ These problems are suited to solution using program *FOURBAR* which is on the attached CD-ROM.

**FIGURE P12-6**

Problem 12-16

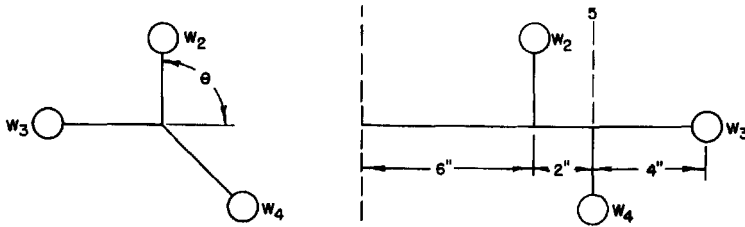


FIGURE P12-7

Problem 12-17

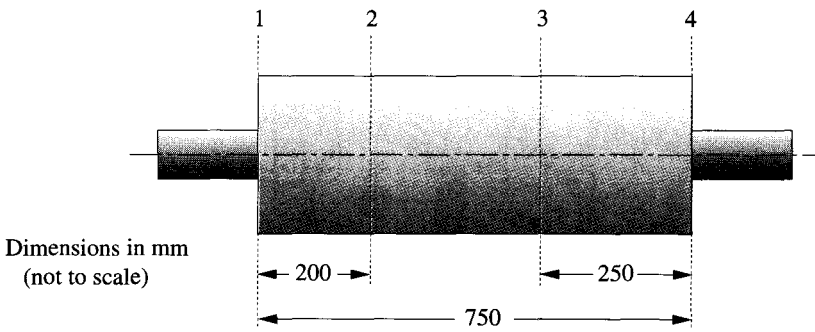
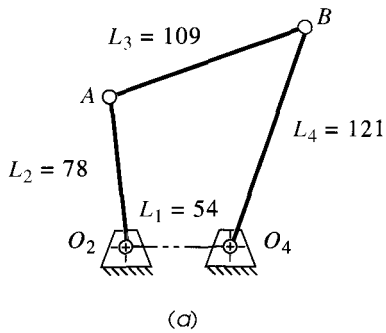


FIGURE P12-8

Problems 12-18 to 12-19



All dimensions in mm

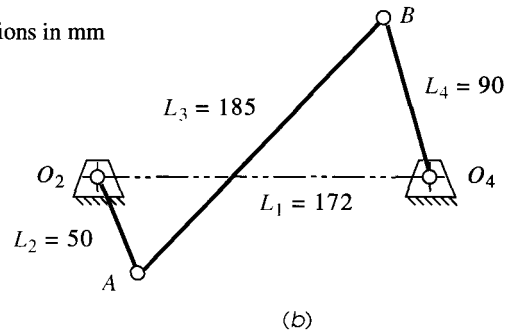
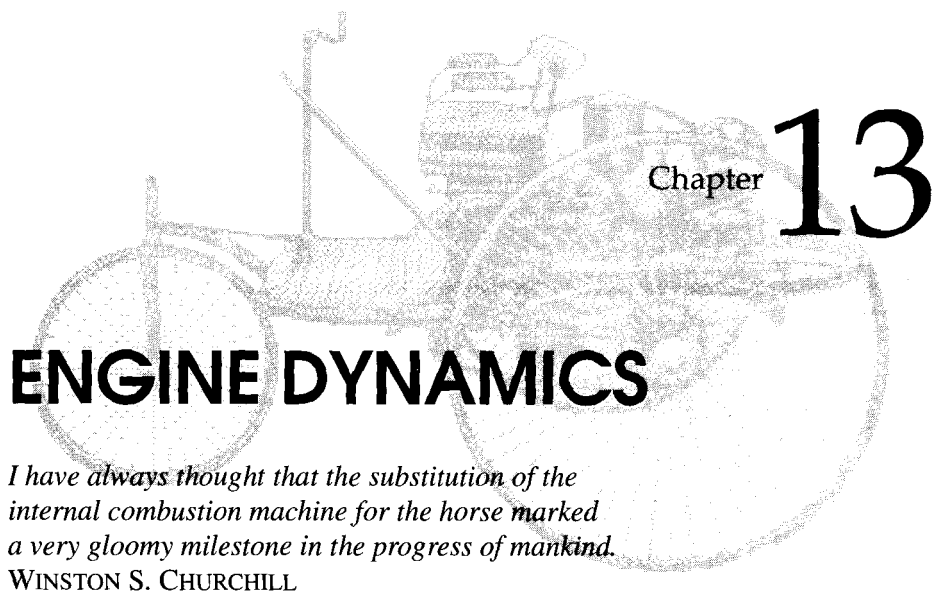


FIGURE P12-9

Problems 12-20 to 12-23



Chapter 13

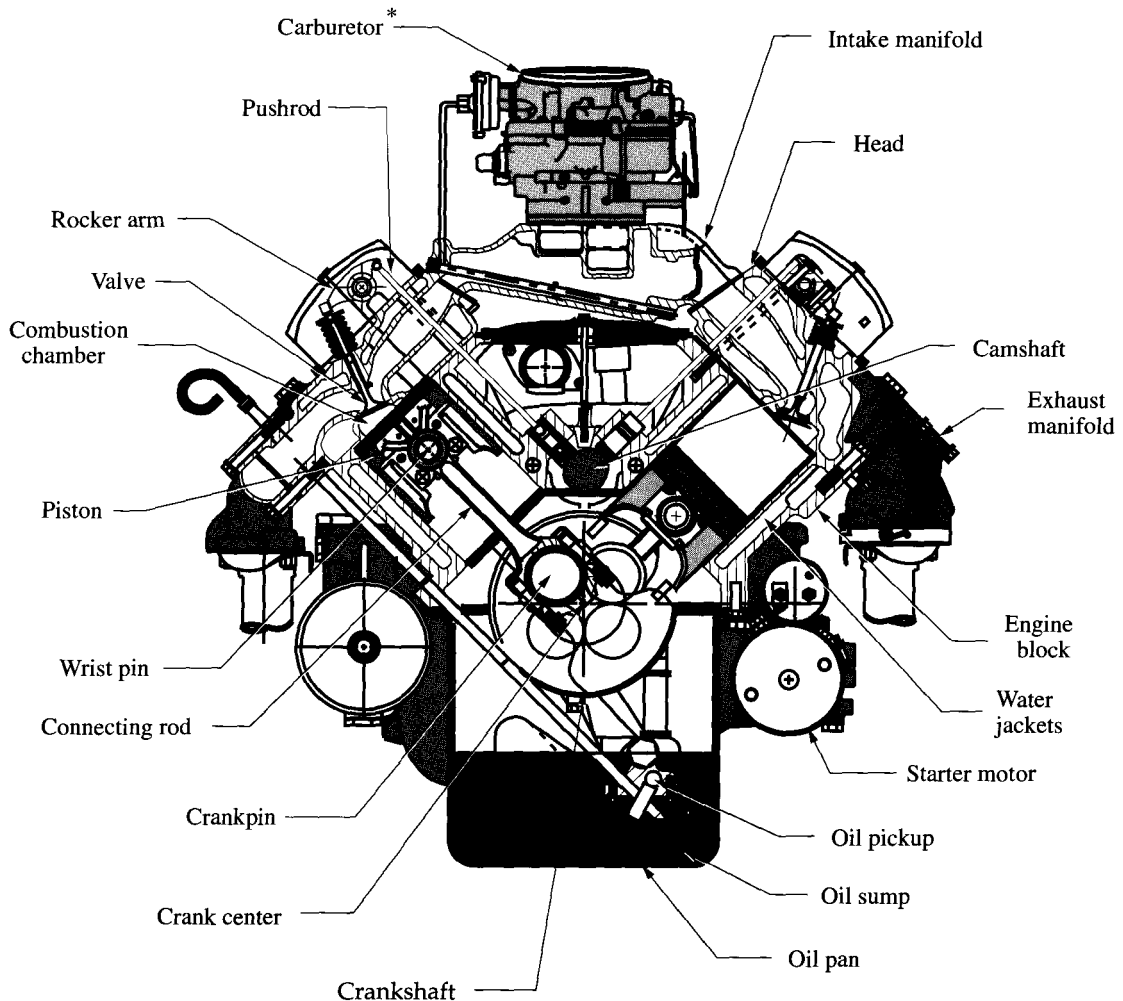
ENGINE DYNAMICS

I have always thought that the substitution of the internal combustion machine for the horse marked a very gloomy milestone in the progress of mankind.
WINSTON S. CHURCHILL

13.0 INTRODUCTION

The previous chapters have introduced analysis techniques for the determination of dynamic forces, moments, and torques in machinery. Shaking forces, moments, and their balancing have also been discussed. We will now attempt to integrate all these dynamic considerations into the design of a common device, the slider-crank linkage as used in the internal combustion engine. This deceptively simple mechanism will be found to be actually quite complex in terms of the dynamic considerations necessary to its design for high-speed operation. Thus it will serve as an excellent example of the application of the dynamics concepts just presented. We will not address the thermodynamic aspects of the internal combustion engine beyond defining the combustion forces which are necessary to drive the device. Many other texts, such as those listed in the bibliography at the end of this chapter, deal with the very complex thermodynamic and fluid dynamic aspects of this ubiquitous device. We will concentrate only on its kinematics and mechanical dynamics aspects. It is not our intention to make an "engine designer" of the student so much as to apply dynamic principles to a realistic design problem of general interest and also to convey the complexity and fascination involved in the design of an apparently simple dynamic device.

Some students may have had the opportunity to disassemble and service an internal combustion engine, but many will have never done so. Thus we will begin with very fundamental descriptions of engine design and operation. The program ENGINE, supplied with this text, is designed to reinforce and amplify the concepts presented. It will perform all the tedious computations necessary to provide the student with dynamic in-

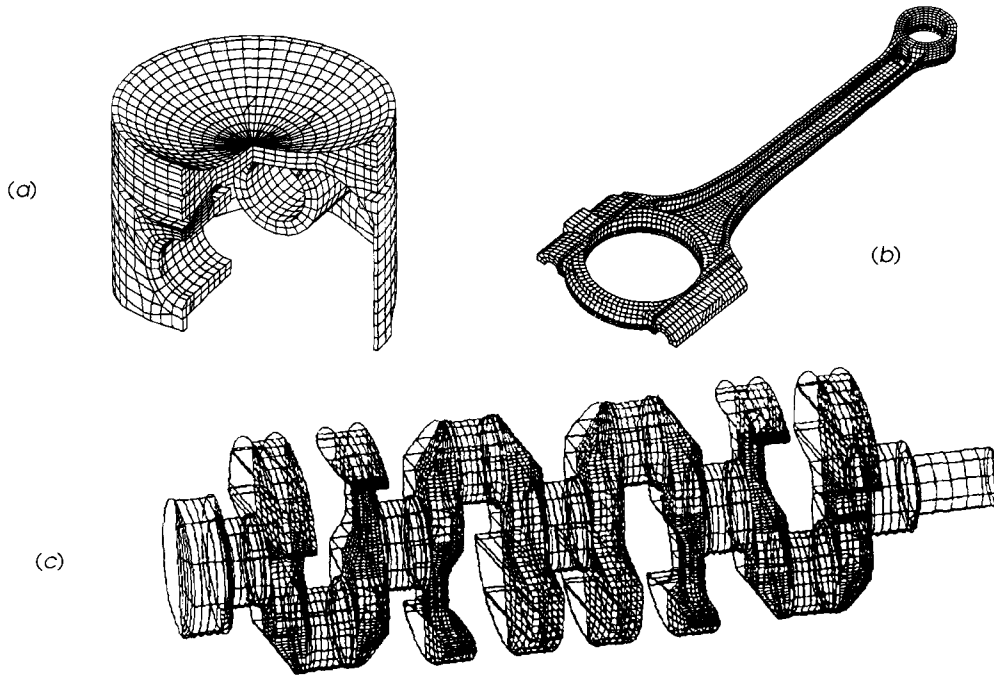
**FIGURE 13-1**

Cutaway cross section of a vee-eight engine

Adapted from a drawing by Lane Thomas, Western Carolina University, Dept. of Industrial Education, with permission.

formation for design choices and trade-offs. The student is encouraged to use this program concurrent with a reading of the text. Many examples and illustrations within the text will be generated with this program and reference will frequently be made to it. A user manual for program ENGINE is provided in Appendix A which can be read or referred to at any time in order to gain familiarity with the program's operation. Examples used in Chapters 13 and 14 which deal with engine dynamics are built into program ENGINE for student observation and exercise. They can be found on a drop-down menu in that program. Other example engine files for program ENGINE are on the CD-ROM.

* Carburetors are being replaced by fuel injection systems on automotive and other engines that are required to meet increasingly stringent exhaust emission control regulations in the United States. Fuel injection gives better control over the fuel-air mixture than a carburetor.

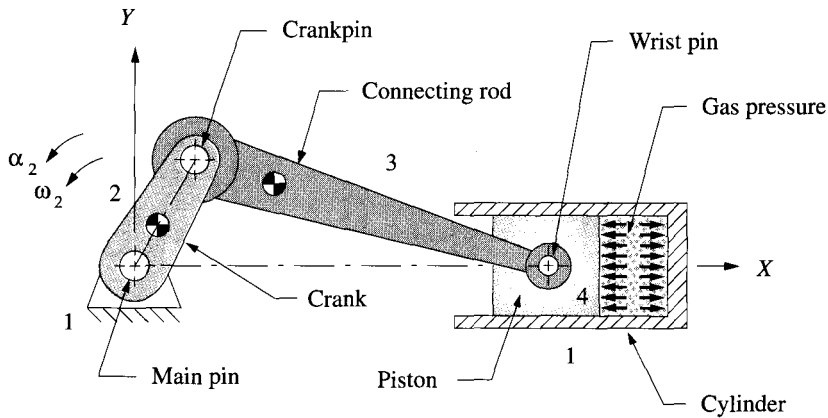
**FIGURE 13-2**

Finite-element models of an engine piston (a), connecting rod (b), and crankshaft (c) *Courtesy of General Motors Co.*

13.1 ENGINE DESIGN

Figure 13-1 (p. 599) shows a detailed cross section of an internal combustion engine. The basic mechanism consists of a crank, a connecting rod (coupler), and piston (slider). Since this figure depicts a **multicylinder vee-eight** engine configuration, there are four cranks arranged on a crankshaft, and eight sets of connecting rods and pistons, four in the left bank of cylinders and four in the right bank. Only two piston-connecting rod assemblies are visible in this view, both on a common crankpin. The others are behind those shown. Figure 13-2 shows finite-element models of a piston, connecting rod and crankshaft for a four-cylinder inline engine. The most usual arrangement is an inline engine with cylinders all in a common plane. Three-, four-, five-, and six-cylinder **inline engines** are in production the world over. **Vee engines** in four-, six-, eight-, ten-, and twelve-cylinder versions are also in production, with vee six and vee eight being the most popular vee configurations. The geometric arrangements of the crankshaft and cylinders have a significant effect on the dynamic condition of the engine. We will explore these effects of multicylinder arrangements in the next chapter. At this stage we wish to deal only with the design of a **single-cylinder** engine. After optimizing the geometry and dynamic condition of one cylinder, we will be ready to assemble combinations of cylinders into multicylinder configurations.

A schematic of the basic **one-cylinder slider-crank** mechanism and the terminology for its principal parts are shown in Figure 13-3. Note that it is "back-driven" com-

**FIGURE 13-3**

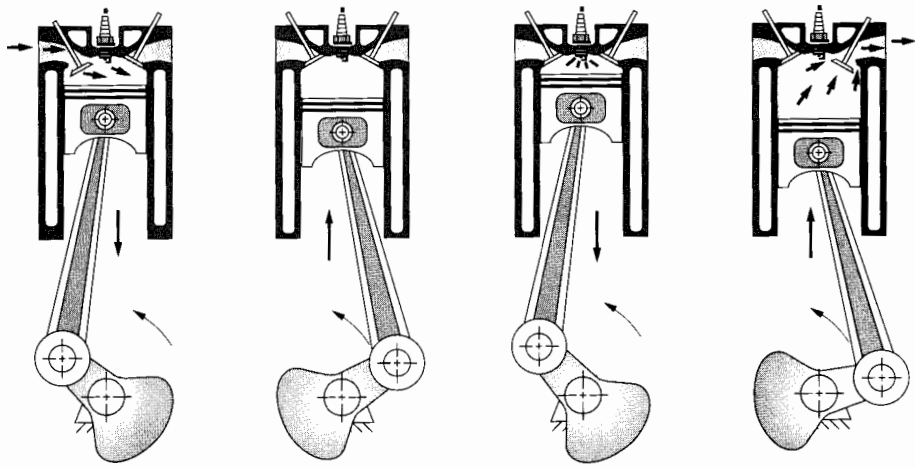
Fourbar slider-crank mechanism for single-cylinder internal combustion engine

pared to the linkages we have been analyzing in previous chapters. That is, the explosion of the combustible mixture in the cylinder drives the piston to the left in Figure 13-3 or down in Figure 13-4, turning the crank. The crank torque that results is ultimately delivered to the drive wheels of the vehicle through a transmission (see Section 9.11, p. 474) to propel the car, motorcycle, or other device. The same slider-crank mechanism can also be used "forward-driven," by motor-driving the crank and taking the output energy from the piston end. It is then called a piston **pump** and is used to compress air, pump gasoline and well water, etc.

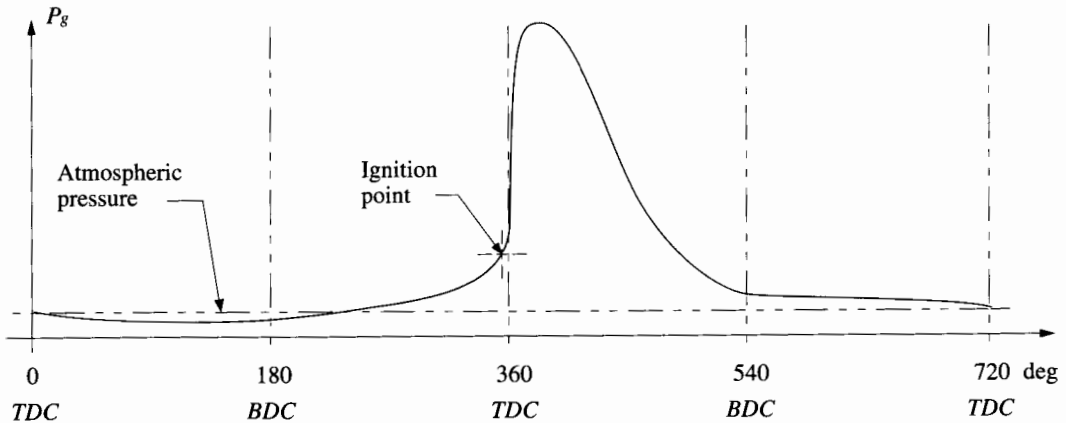
In the internal combustion engine of Figure 13-3, it should be fairly obvious that at most we can only expect energy to be delivered from the exploding gases to the crank during the power stroke of the cycle. The piston must return from bottom dead center (BDC) to top dead center (TDC) on its own momentum before it can receive another push from the next explosion. In fact, some rotational kinetic energy must be stored in the crankshaft merely to carry it through the TDC and BDC points as the moment arm for the gas force at those points is zero. This is why an internal combustion engine must be "spun-up" with a hand crank, pull rope, or starter motor to get it running.

There are two common combustion cycles in use in internal combustion engines, the Clerk two-stroke cycle and the Otto four-stroke cycle, named after their nineteenth century inventors. The four-stroke cycle is most common in automobile, truck, and stationary gasoline engines. The two-stroke cycle is used in motorcycles, outboard motors, chain saws, and other applications where its better power-to-weight ratio outweighs its drawbacks of higher pollution levels and poor fuel economy compared to the four-stroke.

FOUR-STROKE CYCLE The Otto four-stroke cycle is shown in Figure 13-4. It takes four full strokes of the piston to complete one Otto cycle. A piston stroke is defined as its travel from TDC to BDC or the reverse. Thus there are two strokes per 360° crank revolution and it takes 720° of crankshaft rotation to complete one four-stroke cycle. This engine requires at least two valves per cylinder, one for intake and one for exhaust. For discussion, we can start the cycle at any point as it repeats every two crank



(a) Intake stroke (b) Compression stroke (c) Power stroke (d) Exhaust stroke



(e) The gas pressure curve

FIGURE 13-4

The Otto four-stroke combustion cycle

revolutions. Figure 13-4a shows the **intake stroke** which starts with the piston at TDC. A mixture of fuel and air is drawn into the cylinder from the induction system (the fuel injectors, or the carburetor and intake manifold in Figure 13-1, p. 599) as the piston descends to BDC, increasing the volume of the cylinder and creating a slight negative pressure.

During the **compression stroke** in Figure 13-4b, all valves are closed and the gas is compressed as the piston travels from BDC to TDC. Slightly before TDC, a spark is ignited to explode the compressed gas. The pressure from this explosion builds very

quickly and pushes the piston down from TDC to BDC during the power stroke shown in Figure 13-4c. The exhaust valve is opened and the piston's exhaust stroke from BDC to TDC (Figure 13-4d) pushes the spent gases out of the cylinder into the exhaust manifold (see also Figure 13-1) and thence to the catalytic converter for cleaning before being dumped out the tailpipe. The cycle is then ready to repeat with another intake stroke. The valves are opened and closed at the right times in the cycle by a camshaft which is driven in synchrony with the crankshaft by gears, chain, or toothed belt drive. (See Figure 9-24, p. 452.) Figure 13-4e shows the gas pressure curve for one cycle. With a one-cylinder Otto cycle engine, power is delivered to the crankshaft, at most, 25% of the time as there is only 1 power stroke per 2 revolutions.

TWO-STROKE CYCLE The Clerk two-stroke cycle is shown in Figure 13-5. This engine does not need any valves, though to increase its efficiency it is sometimes provided with a passive (pressure differential operated) one at the intake port. It does not have a camshaft or valve train or cam drive gears to add weight and bulk to the engine. As its name implies, it requires only two-strokes, or 360° , to complete its cycle. There is a passageway, called a transfer port, between the combustion chamber above the piston and the crankcase below. There is also an exhaust port in the side of the cylinder. The piston acts to sequentially block or expose these ports as it moves up and down. The crankcase is sealed and mounts the carburetor on it, serving also as the intake manifold.

Starting at TDC (Figure 13-5a), the two-stroke cycle proceeds as follows: The spark plug ignites the fuel-air charge, compressed on the previous revolution. The expansion of the burning gases drives the piston down, delivering torque to the crankshaft. Part-way down, the piston uncovers the exhaust port, allowing the burnt (and also some unburned) gases to begin to escape to the exhaust system.

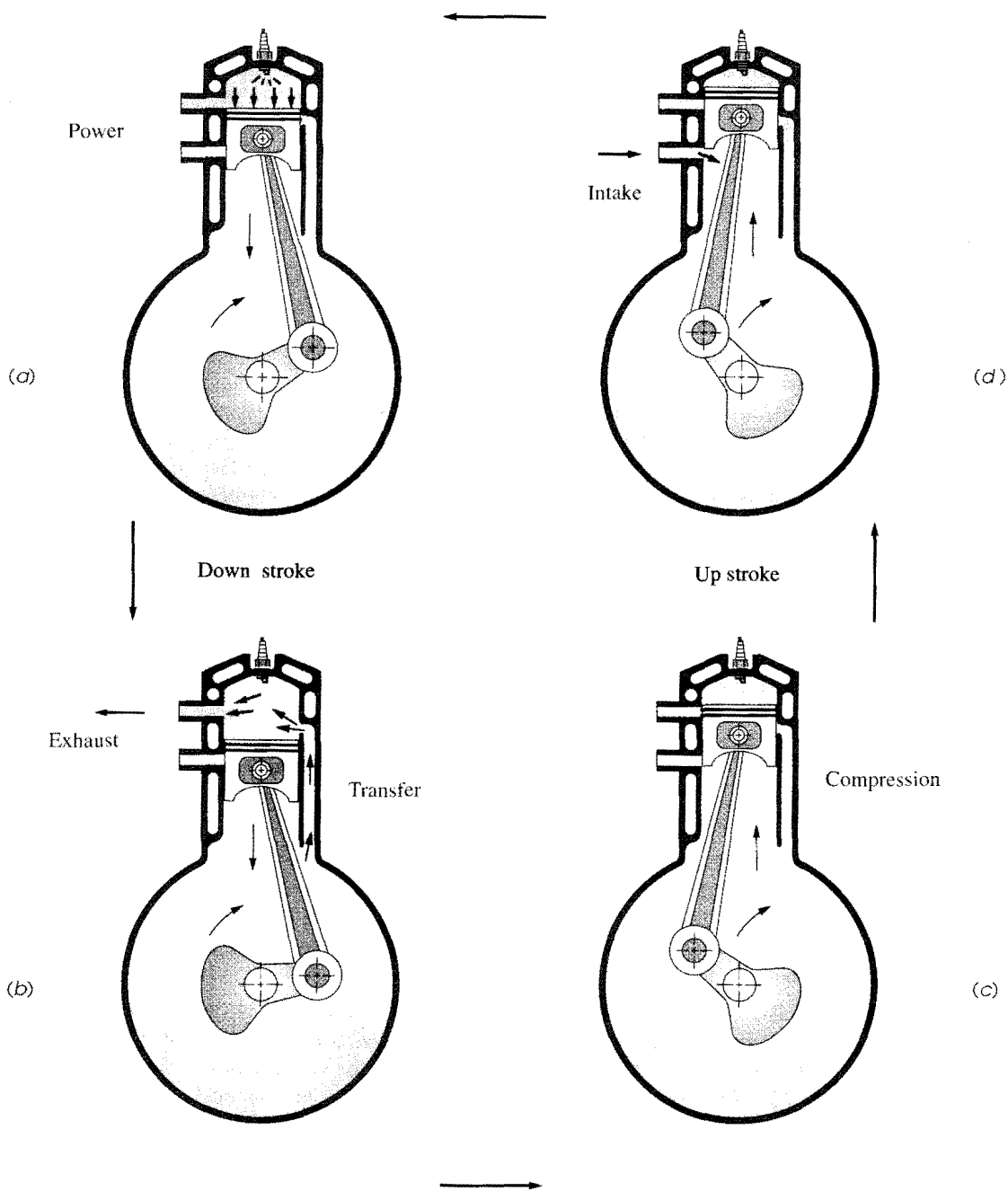
As the piston descends (Figure 13-5ba), it compresses the charge of fuel-air mixture in the sealed crankcase. The piston blocks the intake port preventing blowback through the carburetor. As the piston clears the transfer port in the cylinder wall, its downward motion pushes the new fuel-air charge up through the transfer port to the combustion chamber. The momentum of the exhaust gases leaving the chamber on the other side helps pull in the new charge as well.

The piston passes BDC (Figure 13-5c) and starts up, pushing out the remaining exhaust gases. The exhaust port is closed by the piston as it ascends, allowing compression of the new charge. As the piston approaches TDC, it exposes the intake port (Figure 13-5d), sucking a new charge of air and fuel into the expanded crankcase from the carburetor. Slightly before TDC, the spark is ignited and the cycle repeats as the piston passes TDC.

Clearly, this Clerk cycle is not as efficient as the Otto cycle in which each event is more cleanly separated from the others. Here there is much mixing of the various phases of the cycle. Unburned hydrocarbons are exhausted in larger quantities. This accounts for the poor fuel economy and dirty emissions of the Clerk engine.* It is nevertheless popular in applications where low weight is paramount.

Lubrication is also more difficult in the two-stroke engine than in the four-stroke as the crankcase is not available as an oil sump. Thus the lubricating oil must be mixed with the fuel. This further increases the emissions problem compared to the Otto cycle engine which burns raw gasoline and pumps its lubricating oil separately throughout the engine.

* Research and development is under way to clean up the emissions of the two-stroke engine by using fuel injection and compressed air scavenging of the cylinders. These efforts may yet bring this potentially more powerful engine design into compliance with air quality specifications.

**FIGURE 13-5**

The Clerk two-stroke combustion cycle

DIESEL CYCLE The diesel cycle can be either two-stroke or four-stroke. It is a compression-ignition cycle. No spark is needed to ignite the air-fuel mixture. The air is compressed in the cylinder by a factor of about 14 to 15 (versus 8 to 10 in the spark engine), and a low volatility fuel is injected into the cylinder just before TDC. The heat of compression causes the explosion. Diesel engines are larger and heavier than spark ignition engines for the same power output because the higher pressures and forces at which they operate require stronger, heavier parts. Two-stroke cycle Diesel engines are quite common. Diesel fuel is a better lubricant than gasoline.

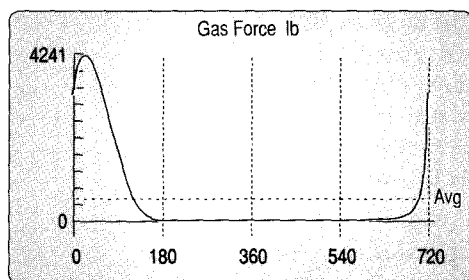
GAS FORCE In all the engines discussed here, the usable output torque is created from the explosive gas pressure generated within the cylinder either once or twice per two revolutions of the crank, depending on the cycle used. The magnitude and shape of this explosion pressure curve will vary with the engine design, stroke cycle, fuel used, speed of operation, and other factors related to the thermodynamics of the system. For our purpose of analyzing the mechanical dynamics of the system, we need to keep the gas pressure function consistent while we vary other design parameters in order to compare the results of our mechanical design changes. For this purpose, program ENGINE has been provided with a built-in gas pressure curve whose peak value is about 600 psi and whose shape is similar to the curve from a real engine. Figure 13-6 shows the gas force curve that results from the built-in gas pressure function in program ENGINE applied to a piston of particular area, for both two- and four-stroke engines. Changes in piston area will obviously affect the gas force magnitude for this consistent pressure function, but no changes in engine design parameters input to this program will change its built-in gas pressure curve. To see this gas force curve, run program ENGINE and select anyone of the example engines from the pull-down menu. Then calculate and plot Gas Force.

13.2 SLIDER-CRANK KINEMATICS

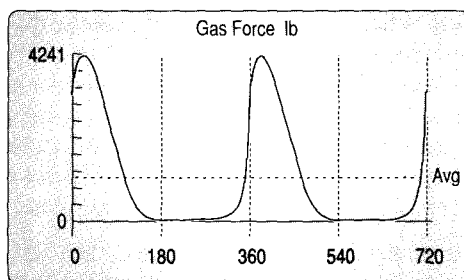
In Chapters 4, 6, 7, and 11 we developed general equations for the exact solution of the positions, velocities, accelerations, and forces in the pin-jointed fourbar linkage, and also for two inversions of the slider-crank linkage, using vector equations. We could again apply that method to the analysis of the "standard" slider-crank linkage, used in the majority of internal combustion engines, as shown in Figure 13-7. Note that its slider motion has been aligned with the X axis. This is a "nonoffset" slider-crank, because the slider axis extended passes through the crank pivot. It also has its slider block translating against the stationary ground plane; thus there will be no Coriolis component of acceleration (see Section 7.3, p. 308).

The simple geometry of this particular inversion of the slider-crank mechanism allows a very straightforward approach to the exact analysis of its slider's position, velocity, and acceleration, using only plane trigonometry and scalar equations. Because of this method's simplicity and to present an alternative solution approach we will analyze this device again.

Let the crank radius be r and the conrod length be l . The angle of the crank is θ , and the angle that the conrod makes with the X axis is ϕ . For any constant crank angular velocity ω , the crank angle $\theta = \omega t$. The instantaneous piston position is x . Two right triangles rqs and lqu are constructed. Then from geometry:



(a) Otto four-stroke cycle



(b) Clerk two-stroke cycle

1 Cylinder
 Bore = 3.00 in
 Stroke = 3.54
 B/S = 0.85
 L/R = 3.50
 RPM = 3400

FIGURE 13-6

Gas force functions in the two-stroke and four-stroke cycle engines

$$q = r \sin \theta = l \sin \phi$$

$$\theta = \omega t$$

(13.1a)

$$\sin \phi = \frac{r}{l} \sin \omega t$$

$$s = r \cos \omega t$$

$$u = l \cos \phi$$

(13.1b)

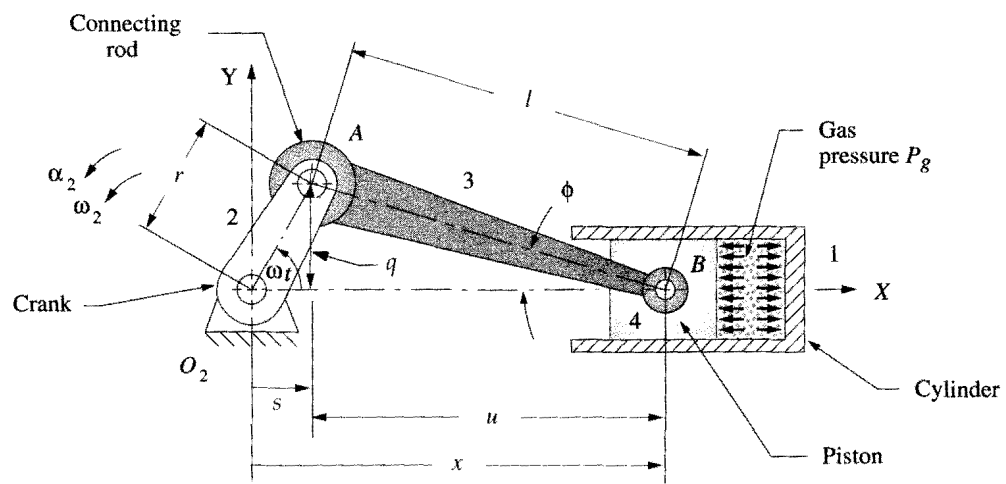
$$x = s + u = r \cos \omega t + l \cos \phi$$

$$\cos \phi = \sqrt{1 - \sin^2 \phi} = \sqrt{1 - \left(\frac{r}{l} \sin \omega t\right)^2} \quad (13.1c)$$

$$x = r \cos \omega t + l \sqrt{1 - \left(\frac{r}{l} \sin \omega t\right)^2} \quad (13.1d)$$

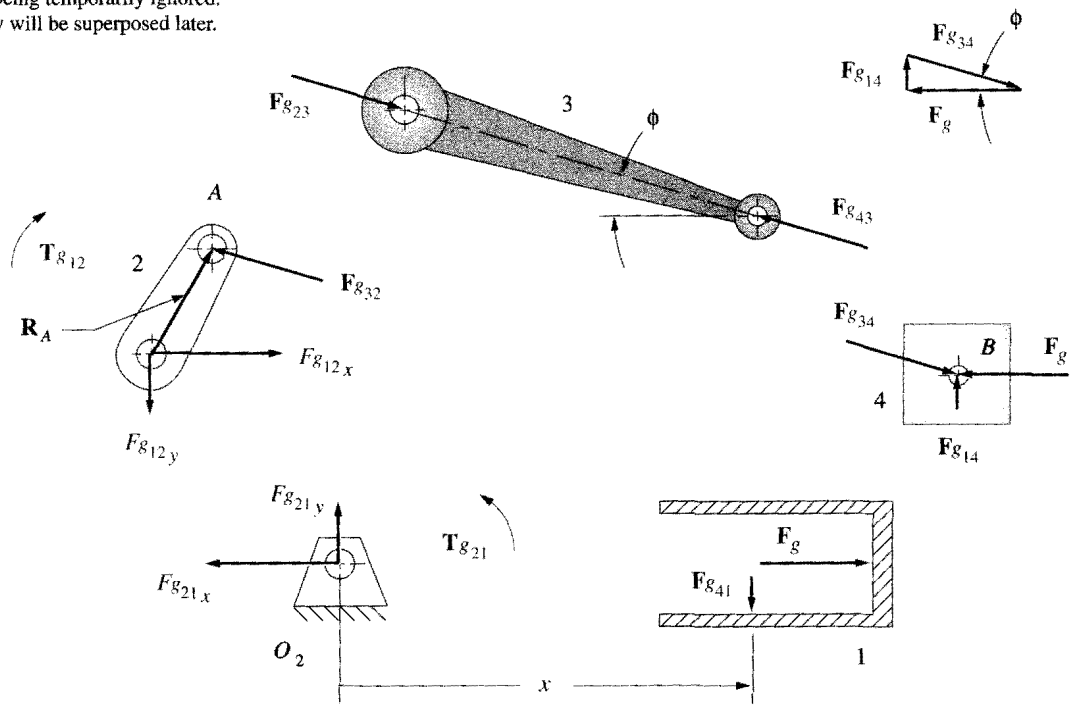
Equation 13.1d is an exact expression for the piston position x as a function of r , l , and ωt . This can be differentiated versus time to obtain exact expressions for the velocity and acceleration of the piston. For a steady-state analysis we will assume ω to be constant.

$$\dot{x} = -r\omega \left[\sin \omega t + \frac{r}{2l} \frac{\sin 2\omega t}{\sqrt{1 - \left(\frac{r}{l} \sin \omega t\right)^2}} \right] \quad (13.1e)$$



(a) The linkage geometry

Note:
Link 3 can be considered as a 2-force member for this gas-force analysis because the inertia forces are being temporarily ignored. They will be superposed later.



(b) Free-body diagrams

FIGURE 13-7
Fourbar slider-crank linkage position and gas force analysis. (See Figure 13-12, p. 618 for the inertia force analysis.)

$$\ddot{x} = -r\omega^2 \left\{ \cos \omega t - \frac{r \left[l^2 (1 - 2 \cos^2 \omega t) - r^2 \sin^4 \omega t \right]}{\left[l^2 - (r \sin \omega t)^2 \right]^{\frac{3}{2}}} \right\} \quad (13.1f)$$

Equations 13.1 can easily be solved with a computer for all values of ωt needed. But, it is rather difficult to look at equation 13.1f and visualize the effects of changes in the design parameters r and l on the acceleration. It would be useful if we could derive a simpler expression, even if approximate, that would allow us to more easily predict the results of design decisions involving these variables. To do so, we will use the binomial theorem to expand the radical in equation 13.1d for piston position to put the equations for position, velocity, and acceleration in simpler, approximate forms which will shed some light on the dynamic behavior of the mechanism.

The general form of the binomial theorem is:

$$(a + b)^n = a^n + na^{n-1}b + \frac{n(n-1)}{2!}a^{n-2}b^2 + \frac{n(n-1)(n-2)}{3!}a^{n-3}b^3 + \dots \quad (13.2a)$$

The radical in equation 13.1d is:

$$\sqrt{1 - \left(\frac{r}{l} \sin \omega t \right)^2} = \left[1 - \left(\frac{r}{l} \sin \omega t \right)^2 \right]^{\frac{1}{2}} \quad (13.2b)$$

where, for the binomial expansion:

$$a = 1 \qquad b = -\left(\frac{r}{l} \sin \omega t \right)^2 \qquad n = \frac{1}{2} \quad (13.2c)$$

It expands to:

$$1 - \frac{1}{2} \left(\frac{r}{l} \sin \omega t \right)^2 + \frac{1}{8} \left(\frac{r}{l} \sin \omega t \right)^4 - \frac{1}{16} \left(\frac{r}{l} \sin \omega t \right)^6 + \dots \quad (13.2d)$$

$$\text{or:} \qquad 1 - \left(\frac{r^2}{2l^2} \right) \sin^2 \omega t + \left(\frac{r^4}{8l^4} \right) \sin^4 \omega t - \left(\frac{r^6}{16l^6} \right) \sin^6 \omega t + \dots \quad (13.2e)$$

Each nonconstant term contains the **crank-conrod ratio** r/l to some power. Applying some engineering common sense to the depiction of the slider-crank in Figure 13-7a (p. 607), we can see that if r/l were greater than 1 the crank could not make a complete revolution. In fact if r/l even gets close to 1, the piston will hit the fixed pivot O_2 before the crank completes its revolution. If r/l is as large as 1/2, the transmission angle ($\pi/2 - \phi$) will be too small (see Sections 3.3, p. 81 and 4.10, p. 169) and the linkage will not run well. A practical upper limit on the value of r/l is about 1/3. Most slider-crank linkages will have this **crank-conrod ratio** somewhere between 1/3 and 1/5 for smooth operation. If we substitute this practical upper limit of $r/l = 1/3$ into equation 13.2e, we get:

$$1 - \left(\frac{1}{18}\right)\sin^2 \omega t + \left(\frac{1}{648}\right)\sin^4 \omega t - \left(\frac{1}{11,664}\right)\sin^6 \omega t + \dots \quad (13.2f)$$

$$1 - 0.05556\sin^2 \omega t + 0.00154\sin^4 \omega t - 0.00009\sin^6 \omega t + \dots$$

Clearly we can drop all terms after the second with very small error. Substituting this approximate expression for the radical in equation 13.1d gives an approximate expression for piston displacement with only a fraction of one percent error.

$$x \cong r \cos \omega t + l \left[1 - \left(\frac{r^2}{2l^2} \right) \sin^2 \omega t \right] \quad (13.3a)$$

Substitute the trigonometric identity:

$$\sin^2 \omega t = \frac{1 - \cos 2\omega t}{2} \quad (13.3b)$$

and simplify:

$$x \cong l - \frac{r^2}{4l} + r \left(\cos \omega t + \frac{r}{4l} \cos 2\omega t \right) \quad (13.3c)$$

Differentiate for velocity of the piston (with constant ω):

$$\dot{x} \cong -r\omega \left(\sin \omega t + \frac{r}{2l} \sin 2\omega t \right) \quad (13.3d)$$

Differentiate again for acceleration (with constant ω):

$$\ddot{x} \cong -r\omega^2 \left(\cos \omega t + \frac{r}{l} \cos 2\omega t \right) \quad (13.3e)$$

The process of binomial expansion has, in this particular case, led us to Fourier series approximations of the exact expressions for the piston displacement, velocity, and acceleration. Fourier* showed that any periodic function can be approximated by a series of sine and cosine terms of integer multiples of the independent variable. Recall that we dropped the fourth, sixth, and subsequent power terms from the binomial expansion, which would have provided $\cos 400t$, $\cos 600t$, etc., terms in this expression. These multiple angle functions are referred to as the **harmonics** of the fundamental $\cos \omega t$ term. The $\cos \omega t$ term repeats once per crank revolution and is called the fundamental frequency or the **primary component**. The second harmonic, $\cos 2\omega t$, repeats twice per crank revolution and is called the **secondary component**. The higher harmonics were dropped when we truncated the series. The constant term in the displacement function is the **DC component or average value**. The complete function is the sum of its harmonics. The Fourier series form of the expressions for displacement and its derivatives lets us see the relative contributions of the various harmonic components of the functions. This approach will prove to be quite valuable when we attempt to dynamically balance an engine design.

Program ENGINE calculates the position, velocity, and acceleration of the piston according to equations 13.3c, d, and e. Figure 13-8a, b, and c shows these functions for

* Baron Jean Baptiste Joseph Fourier (1768-1830) published the description of the mathematical series which bears his name in *The Analytic Theory of Heat* in 1822. The Fourier series is widely used in harmonic analysis of all types of physical systems. Its general form is:

$$y = \frac{a_0}{2} + (a_1 \cos x + b_1 \sin x) + (a_2 \cos 2x + b_2 \sin 2x) + \dots + (a_n \cos nx + b_n \sin nx)$$

this example engine in the program as plotted for constant crank ω over two full revolutions. The acceleration curve shows the effects of the second harmonic term most clearly because that term's coefficient is larger than its correspondent in either of the other two functions. The fundamental ($-\cos \omega t$) term gives a pure harmonic function with a period of 360° . This fundamental term dominates the function as it has the largest coefficient in equation 13.3e. The flat top and slight dip in the positive peak acceleration of Figure 13-8c is caused by the $\cos 2\omega t$ second harmonic adding or subtracting from the fundamental. Note the very high value of peak acceleration of the piston even at the midrange engine speed of 3400 rpm. It is $747 g$'s! At 6000 rpm this increases to nearly $1300 g$'s. This is a moderately sized engine, of 3-in (76 mm) bore and 3.54-in (89 mm) stroke, with 25-in^3 (400-cc) displacement per cylinder.

SUPERPOSITION We will now analyze the dynamic behavior of the single-cylinder engine based on the approximate kinematic model developed above. Since we have several sources of dynamic excitation to deal with, we will use the method of superposition to separately analyze them and then combine their effects. We will first consider the **forces and torques** which are due to the presence of the **explosive gas forces** in the cylinder, which drive the engine. Then we will analyze the **inertia forces and torques** that result from the high-speed motion of the elements. The total force and torque state of the machine at any instant will be the sum of these components. Finally we will look at the **shaking forces and torques** on the ground plane and the **pin forces** within the linkage that result from the combination of applied and dynamic forces on the system.

13.3 GAS FORCE AND GAS TORQUE

The **gas force** is due to the gas pressure from the exploding fuel-air mixture impinging on the top of the piston surface as shown in Figure 13-3 (p. 601). Let F_g = gas force, P_g = gas pressure, A_p = area of piston, and B = bore of cylinder, which is also equal to the piston diameter. Then:

$$\begin{aligned} \mathbf{F}_g &= -P_g A_p \hat{\mathbf{i}}; & A_p &= \frac{\pi}{4} B^2 \\ \mathbf{F}_g &= -\frac{\pi}{4} P_g B^2 \hat{\mathbf{i}} \end{aligned} \quad (13.4)$$

The negative sign is due to the choice of engine orientation in the coordinate system of Figure 13-3. The gas pressure P_g in this expression is a function of crank angle θ and is defined by the thermodynamics of the engine. A typical gas pressure curve for a four-stroke engine is shown in Figure 13-4 (p. 602). The gas force curve shape is identical to that of the gas pressure curve as they differ only by a constant multiplier, the piston area A_p . Figure 13-6 (p. 606) shows the approximation of the gas force curve used in program ENGINE for both four- and two-stroke engines.

The **gas torque** in Figure 13-9 is due to the gas force acting at a moment arm about the crank center O2 in Figure 13-7 (p. 607). This moment arm varies from zero to a maximum as the crank rotates. The distributed gas force over the piston surface has been resolved to a single force acting through the mass center of link 4 in the free-body diagrams of Figure 13-7b. The concurrent force system at point B is resolved in the vector diagram showing that:

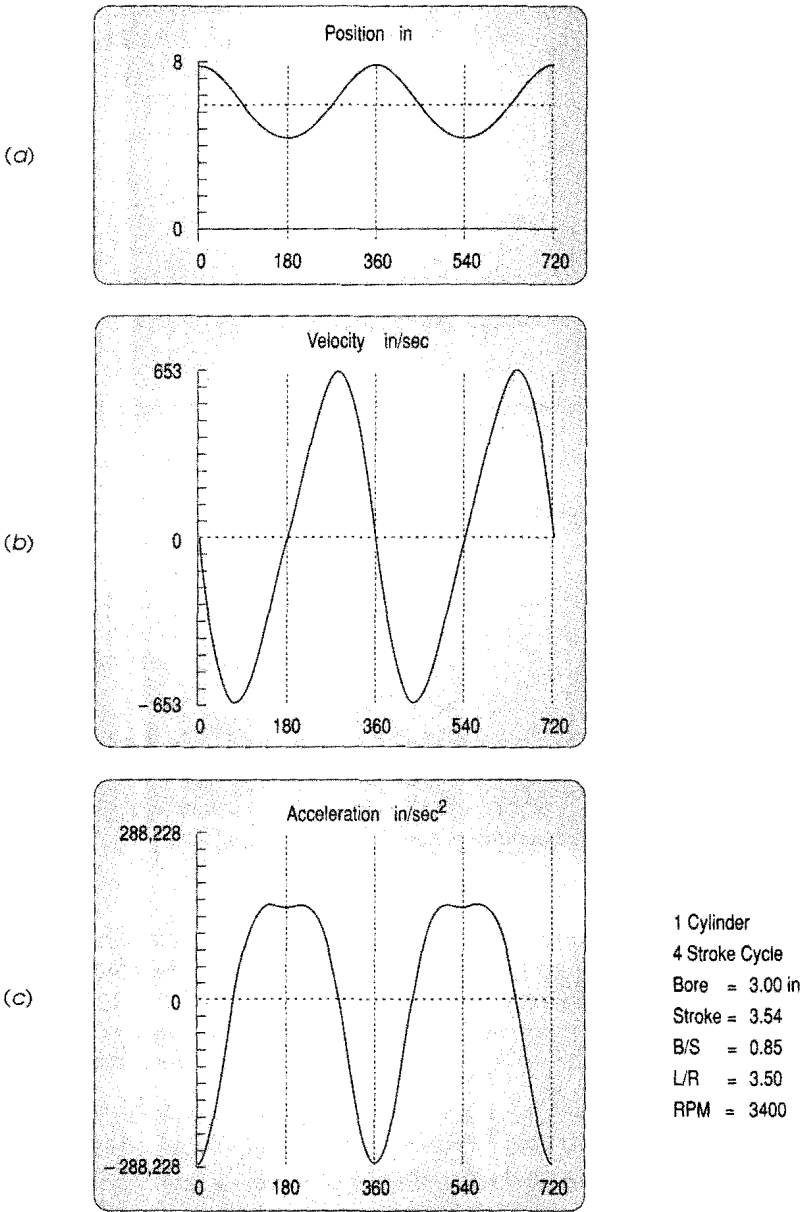
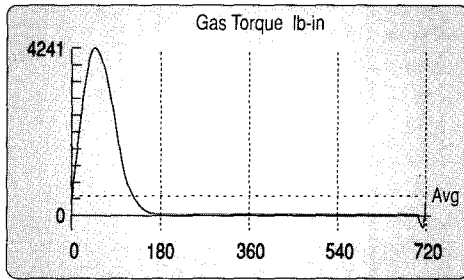


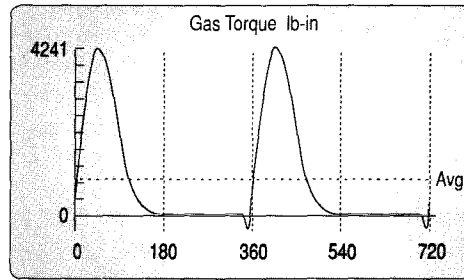
FIGURE 13-8
Position, velocity, and acceleration functions for a single-cylinder engine

$$F_{g14} = F_g \tan \phi \hat{j} \tag{13.5a}$$

$$F_{g34} = F_g \hat{i} - F_g \tan \phi \hat{j} \tag{13.5b}$$



(a) Otto four-stroke cycle



(b) Clerk two-stroke cycle

1 Cylinder
 Bore = 3.00 in
 Stroke = 3.54
 B/S = 0.85
 L/R = 3.50
 RPM = 3400

FIGURE 13-9

Gas torque functions in the two-stroke and four-stroke cycle engines

From the free-body diagrams in Figure 13-7 we can see that:

$$\mathbf{F}_{g41} = -\mathbf{F}_{g14}$$

$$\mathbf{F}_{g43} = -\mathbf{F}_{g34}$$

$$\mathbf{F}_{g23} = -\mathbf{F}_{g43}$$

$$\mathbf{F}_{g32} = -\mathbf{F}_{g23}$$

so:

$$\mathbf{F}_{g32} = -\mathbf{F}_{g34} = -F_g \hat{\mathbf{i}} + F_g \tan \phi \hat{\mathbf{j}} \quad (13.5c)$$

The **driving torque** $\mathbf{T}_{g21}$ at link 2 due to the gas force can be found from the cross product of the position vector to point A and the force at point A.

$$\mathbf{T}_{g21} = \mathbf{R}_A \times \mathbf{F}_{g32} \quad (13.6a)$$

This expression can be expanded and will involve the crank length r and the angles θ and ϕ as well as the gas force $\mathbf{F}_g$. Note from the free-body diagram for link 1 that we can also express the torque in terms of the forces $\mathbf{F}_{g14}$ or $\mathbf{F}_{g41}$ which act always perpendicular to the motion of the slider (neglecting friction), and the distance x , which is their instantaneous moment arm about O_2 . The reaction torque $\mathbf{T}_{g12}$ due to the gas force trying to rock the ground plane is:

$$\mathbf{T}_{g12} = F_{g41} \cdot x \hat{\mathbf{k}} \quad (13.6b)$$

If you have ever abruptly opened the throttle of a running automobile engine while working on it, you probably noticed the engine move to the side as it rocked in its mounts from the reaction torque. The driving torque $\mathbf{T}_{g21}$ is the negative of this reaction torque.

$$\mathbf{T}_{g21} = -\mathbf{T}_{g12}$$

$$\mathbf{T}_{g21} = -F_{g41} \cdot x \hat{\mathbf{k}} \quad (13.6c)$$

and:

$$F_{g14} = -F_{g41}$$

so:

$$\mathbf{T}_{g21} = F_{g14} \cdot x \hat{\mathbf{k}} \quad (13.6d)$$

Equation 13.6d gives us an expression for **gas torque** which involves the displacement of the piston x for which we have already derived equation 13.3a. Substituting equation 13.3a for x and the magnitude of equation 13.5a for F_{g14} we get:

$$\mathbf{T}_{g21} = (F_g \tan \phi) \left[l - \frac{r^2}{4l} + r \left(\cos \omega t + \frac{r}{4l} \cos 2\omega t \right) \right] \hat{\mathbf{k}} \quad (13.6e)$$

Equation 13.6e contains the conrod angle ϕ as well as the independent variable, crank angle ωt . We would like to have an expression which involves only ωt . We can substitute an expression for $\tan \phi$ generated from the geometry of Figure 13-7a.

$$\tan \phi = \frac{q}{u} = \frac{r \sin \omega t}{l \cos \phi} \quad (13.7a)$$

Substitute equation 13.1c for $\cos \phi$:

$$\tan \phi = \frac{r \sin \omega t}{l \sqrt{1 - \left(\frac{r}{l} \sin \omega t \right)^2}} \quad (13.7b)$$

The radical in the denominator can be expanded using the binomial theorem as was done in equations 13.2 (p. 608), and the first two terms retained for a good approximation to the exact expression,

$$\frac{1}{\sqrt{1 - \left(\frac{r}{l} \sin \omega t \right)^2}} \cong 1 + \frac{r^2}{2l^2} \sin^2 \omega t \quad (13.7c)$$

giving:

$$\tan \phi \cong \frac{r}{l} \sin \omega t \left(1 + \frac{r^2}{2l^2} \sin^2 \omega t \right) \quad (13.7d)$$

Substitute this into equation 13.6e for the gas torque:

$$\mathbf{T}_{g21} \cong F_g \left[\frac{r}{l} \sin \omega t \left(1 + \frac{r^2}{2l^2} \sin^2 \omega t \right) \right] \left[l - \frac{r^2}{4l} + r \left(\cos \omega t + \frac{r}{4l} \cos 2\omega t \right) \right] \hat{\mathbf{k}} \quad (13.8a)$$

Expand this expression and neglect any terms containing the conrod crank ratio r/l raised to any power greater than one since these will have very small coefficients as was seen in equation 13.2. This results in a simpler, but even more approximate expression for the gas torque:

$$\mathbf{T}_{g21} \cong F_g r \sin \omega t \left(1 + \frac{r}{l} \cos \omega t \right) \hat{\mathbf{k}} \quad (13.8b)$$

Note that the **exact value** of this **gas torque** can always be calculated from equations 13.1d, 13.5a and 13.6d in combination, or from the expansion of equation 13.6a, if you require a more accurate answer. For design purposes the approximate equation 13.8b will usually be adequate. Program ENGINE calculates the gas torque using equation 13.8b and its built-in gas pressure curve to generate the gas force function. Plots of the gas torque for two- and four-stroke cycles are shown in Figure 13-9. Note the simi-

larity in shape to that of the gas force curve in Figure 13-6 (p. 606). Note also that the two-stroke has theoretically twice the power available as the four-stroke, all other factors equal, because there are twice as many torque pulses per unit time. The poorer efficiency of the two-stroke reduces this theoretical advantage, however.

13.4 EQUIVALENT MASSES

To do a complete dynamic force analysis on any mechanism we need to know the geometric properties (mass, center of gravity, mass moment of inertia) of the moving links as was discussed in previous chapters (see Sections 10.3 to 10.8 and Chapter 11). This is easy to do if the link already is designed in detail and its dimensions are known. When designing the mechanism from scratch, we typically do not yet know that level of detail about the links' geometries. But we must nevertheless make some estimate of their geometric parameters in order to begin the iteration process which will eventually converge on a detailed design.

In the case of this slider-crank mechanism, the **crank** is in **pure rotation** and the **piston** is in **pure translation**. By assuming some reasonable geometries and materials we can make approximations of their dynamic parameters. Their kinematic motions are easily determined. We have already derived expressions for the piston motion in equations 13.3 (p. 609). Further, if we balance the rotating crank, as described and recommended in the previous chapter, then the CG of the crank will be motionless at its center *O2* and will not contribute to the dynamic forces. We will do this in a later section.

The conrod is in complex motion. To do an exact dynamic analysis as was derived in Section 11.5, we need to determine the linear acceleration of its CG for all positions. At the outset of the design, the conrod's CG location is not accurately defined. To "bootstrap" the design, we need a simplified model of this connecting rod which we can later refine, as more dynamic information is generated about our engine design. The requirements for a dynamically equivalent model were stated in Section 10.2 and are repeated here as Table 13-1 for your convenience.

If we could model our still-to-be-designed conrod as two, lumped, point masses, concentrated one at the crankpin (point *A* in Figure 13-7, p. 607), and one at the wrist pin (point *B* in Figure 13-7), we would at least know what the motions of these lumps are. The lump at *A* would be in pure rotation as part of the crank, and the lump at point *B* would be in pure translation as part of the piston. These lumped, point masses have no dimension and are assumed to be connected with a magical, massless but rigid rod.*

DYNAMICALLY EQUIVALENT MODEL Figure 13-10a shows a typical conrod. Figure 13-10b shows a generic two-mass model of the conrod. One mass m_t is located at distance l_t from the CG of the original rod, and the second mass m_p at distance l_p from the CG. The mass of the original part is m_3 , and its moment of inertia about its CG is I_{G3} . Expressing the three requirements for dynamic equivalence from Table 13-1 mathematically in terms of these variables, we get:

$$m_p + m_t = m_3 \quad (13.9a)$$

$$m_p l_p = m_t l_t \quad (13.9b)$$

$$m_p l_p^2 + m_t l_t^2 = I_{G3} \quad (13.9c)$$

* These lumped mass models have to be made with very special materials. *Unobtainium 206* has the property of infinite mass density, thus occupies no space and can be used for "point masses." *Unobtainium 208* has infinite stiffness and zero mass and thus can be used for rigid but "massless rods."

There are four unknowns in these three equations, m_p , l_p , m_b , l_b , which means we must choose a value for any one variable to solve the system. Let us choose the distance l_t equal to the distance to the wrist pin, l_b , as shown in Figure 13-10c. This will put one mass at a desired location. Solving equations 13.9a and 13.9b simultaneously with that substitution gives expressions for the two lumped masses:

$$\begin{aligned} m_p &= m_3 \frac{l_b}{l_p + l_b} \\ m_b &= m_3 \frac{l_p}{l_p + l_b} \end{aligned} \quad (13.9d)$$

Substituting equation 13.9d into 13.9c gives a relation between l_p and l_b :

$$\begin{aligned} m_3 \frac{l_b}{l_p + l_b} l_p^2 + m_3 \frac{l_p}{l_p + l_b} l_b^2 &= I_{G_3} = m_3 l_p l_b \\ l_p &= \frac{I_{G_3}}{m_3 l_b} \end{aligned} \quad (13.9e)$$

Please refer to Section 10.8 and equation 10.13 which define the *center of percussion* and its geometric relationship to a corresponding *center of rotation*. Equation 13.9e is the same as equation 11.13 (except for sign which is due to an arbitrary choice of the link's orientation in the coordinate system). The distance l_p is the location of the center of percussion corresponding to a center of rotation at l_b . Thus our second mass m_p must be placed at the link's **center of percussion** P (using point B as its center of rotation) to obtain exact dynamic equivalence. The masses must be as defined in equation 13.9d.

The geometry of the typical conrod, as shown in Figures 13-2 (p. 600) and 13-10a, is large at the crankpin end (A) and small at the wrist pin end (B). This puts the CG close to the "big end." The center of percussion P will be even closer to the big end than is the CG . For this reason, we can place the second lumped mass, which belongs at P , at point A with relatively small error in our dynamic model's accuracy. This approximate model is adequate for our initial design calculations. Once a viable design geometry is established, we will have to do a complete and exact force analysis with the methods of Chapter 11 before considering the design complete.

Making this substitution of distance l_a for l_p and renaming the lumped masses at those distances m_{3a} and m_{3b} , to reflect both their identity with link 3 and with points A and B , we rewrite equations 13.9d.

TABLE 13-1 Requirements for Dynamic Equivalence

- 1 The mass of the model must equal that of the original body.
- 2 The center of gravity must be in the same location as that of the original body.
- 3 The mass moment of inertia must equal that of the original body.

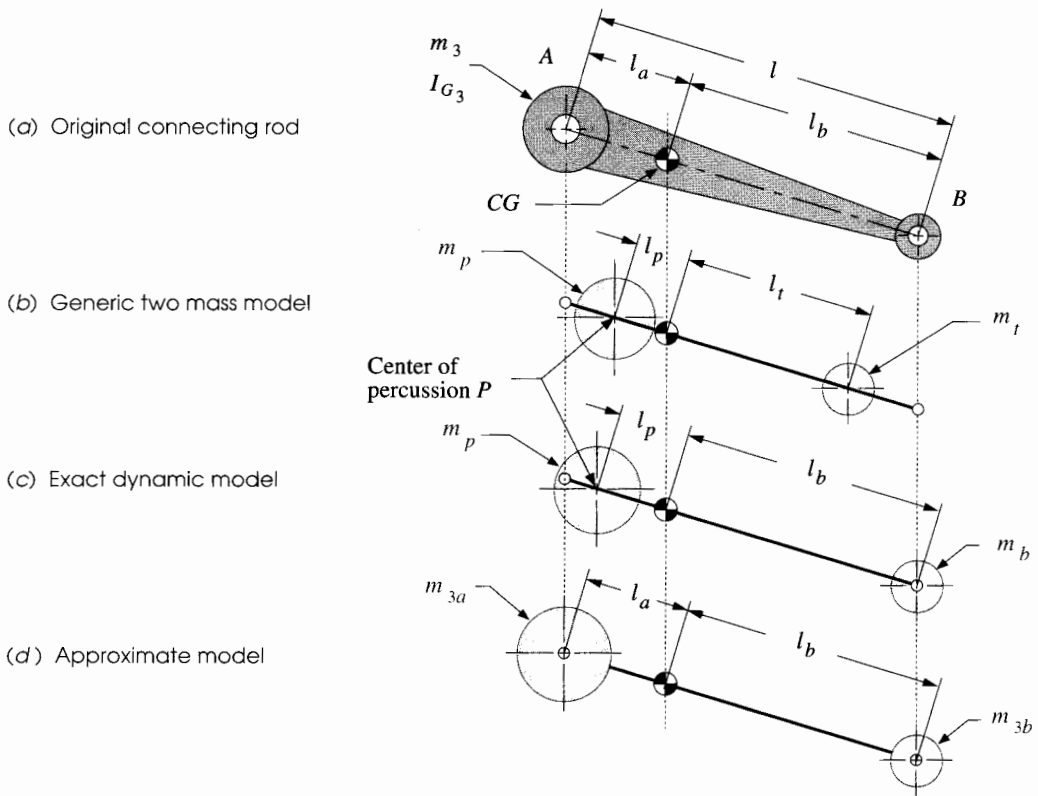


FIGURE 13-10

Lumped mass dynamic models of a connecting rod

let

$$l_p = l_a$$

then :

$$m_{3a} = m_3 \frac{l_b}{l_a + l_b} \quad (13.10a)$$

and :

$$m_{3b} = m_3 \frac{l_a}{l_a + l_b} \quad (13.10b)$$

These define the amounts of the total conrod mass to be placed at each end, to approximately dynamically model that link. Figure 13-10d shows this dynamic model. In the absence of any data on the shape of the conrod at the outset of a design, preliminary dynamic force information can be obtained by using the rule of thumb of placing two-thirds of the conrod's mass at the crankpin and one-third at the wrist pin.

STATICALLY EQUIVALENT MODEL We can create a similar lumped mass model of the crank. Even though we intend to balance the crank before we are done, for generality we will initially model it *unbalanced* as shown in Figure 13-11. Its CG is located at some distance r_{G2} from the pivot, O_2 , on the line to the crankpin, A. We would like to

TABLE 13-2 Requirements for Static Equivalence

- 1 The mass of the model must equal that of the original body.
- 2 The center of gravity must be in the same location as that of the original body.

model it as a lumped mass at A on a massless rod pivoted at O_2 . If our principal concern is with a steady-state analysis, then the crank velocity ω will be held constant. An absence of angular acceleration on the crank allows a statically equivalent model to be used because the equation $\mathbf{T} = I\alpha$ will be zero regardless of the value of I . A **statically equivalent model** needs only to have equivalent mass and equivalent first moments as shown in Table 13-2. The moments of inertia need not match. We model it as two lumped masses, one at point A and one at the fixed pivot O_2 . Writing the two requirements for static equivalence from Table 13-2:

$$\begin{aligned} m_2 &= m_{2a} + m_{2O_2} \\ m_{2a}r &= m_2 r_{G_2} \\ m_{2a} &= m_2 \frac{r_{G_2}}{r} \end{aligned} \quad (13.11)$$

The lumped mass m_{2a} can be placed at point A to represent the unbalanced crank. The second lumped mass, at the fixed pivot O_2 , is not necessary to any calculations as that point is stationary.

These simplifications lead to the lumped parameter model of the slider-crank linkage shown in Figure 13-12. The crankpin, point A , has two masses concentrated at it, the equivalent mass of the crank m_{2a} and the portion of conrod m_{3a} . Their sum is m_A . At the wrist pin, point B , two masses are also concentrated, the piston mass m_4 and the remaining portion of the conrod mass m_{3b} . Their sum is m_B . This model has masses which are either in pure rotation (m_A) or in pure translation (m_B), so it is very easy to dynamically analyze.

$$\begin{aligned} m_A &= m_{2a} + m_{3a} \\ m_B &= m_{3b} + m_4 \end{aligned} \quad (13.12)$$

VALUE OF MODELS *The value of constructing simple, lumped mass models of complex systems increases with the complexity of the system being designed.* It makes little sense to spend large amounts of time doing sophisticated, detailed analyses of designs which are so ill-defined at the outset that their conceptual viability is as yet unproven. Better to get a reasonably approximate and rapid answer that tells you the concept needs to be rethought, than to spend a greater amount of time reaching the same conclusion to more decimal places.

13.5 INERTIA AND SHAKING FORCES

The simplified, lumped mass model of Figure 13-12 can be used to develop expressions for the forces and torques due to the accelerations of the masses in the system. The method of d'Alembert is of value in visualizing the effects of these moving masses on the sys-

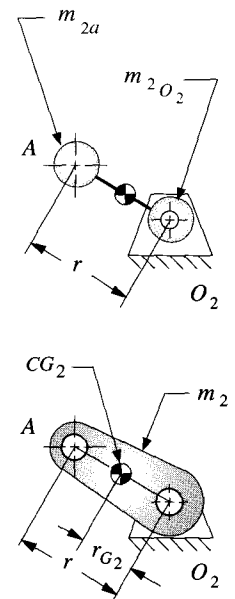


FIGURE 13-11
Statically equivalent
lumped mass model of
a crank

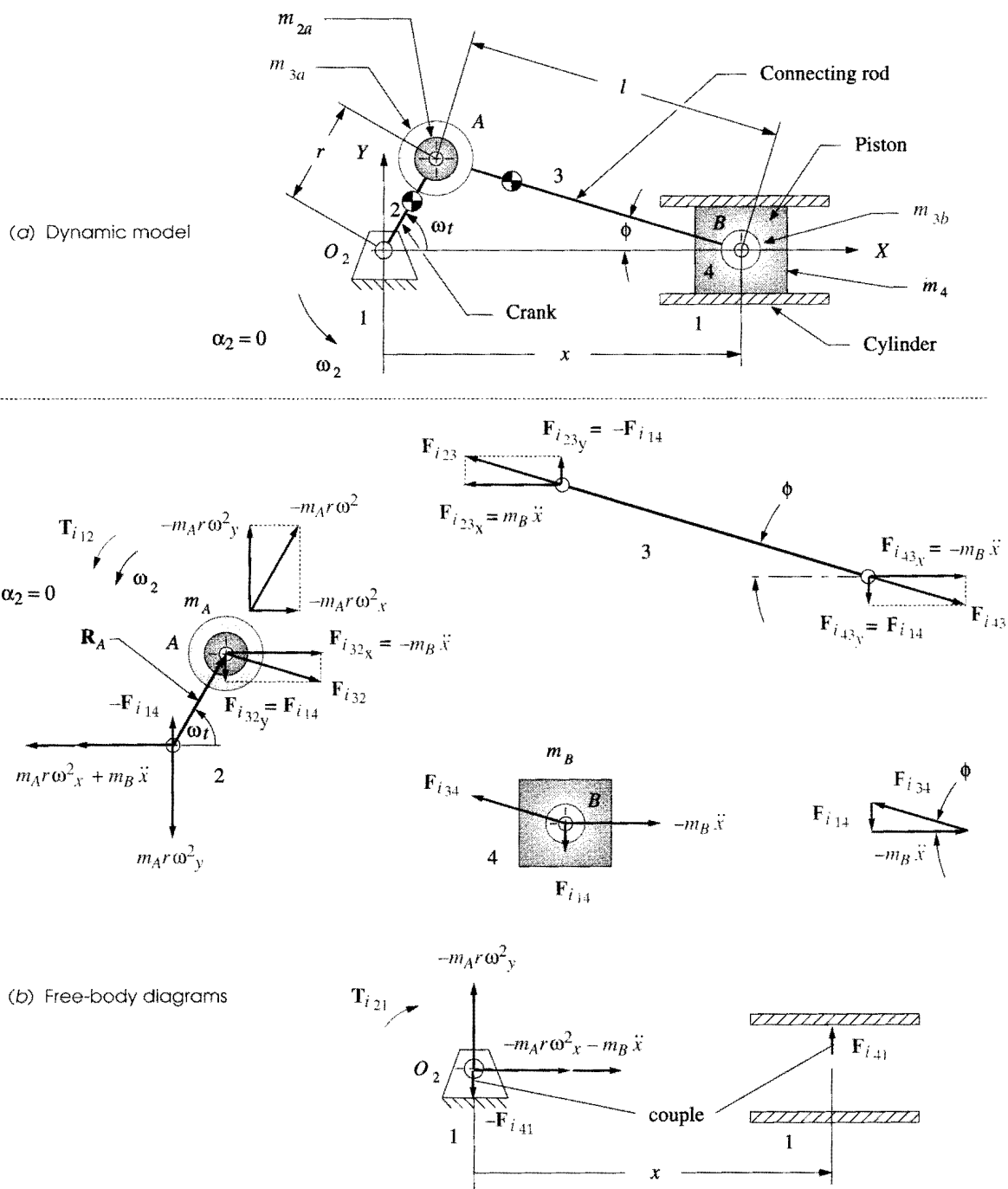


FIGURE 13-12

Lumped mass dynamic model of the slider-crank

tem and on the ground plane. Accordingly, the free-body diagrams of Figure 13-12b show the d'Alembert inertia forces acting on the masses at points A and B . Friction is again ignored. The acceleration for point B is given in equation 13.3e. The acceleration of point A in pure rotation is obtained by differentiating the position vector $\mathbf{R}_A$ twice, assuming a constant crankshaft ω , which gives:

$$\begin{aligned}\mathbf{R}_A &= r \cos \omega t \hat{\mathbf{i}} + r \sin \omega t \hat{\mathbf{j}} \\ \mathbf{a}_A &= -r\omega^2 \cos \omega t \hat{\mathbf{i}} - r\omega^2 \sin \omega t \hat{\mathbf{j}}\end{aligned}\quad (13.13)$$

The total inertia force $\mathbf{F}_i$ is the sum of the centrifugal (inertia) force at point A and the inertia force at point B .

$$\mathbf{F}_i = -m_A \mathbf{a}_A - m_B \mathbf{a}_B \quad (13.14a)$$

Breaking it into x and y components:

$$F_{ix} = -m_A (-r\omega^2 \cos \omega t) - m_B \ddot{x} \quad (13.14b)$$

$$F_{iy} = -m_A (-r\omega^2 \sin \omega t) \quad (13.14c)$$

Note that only the x component is affected by the acceleration of the piston. Substituting equation 13.3e into equation 13.14b:

$$\begin{aligned}F_{ix} &\equiv -m_A (-r\omega^2 \cos \omega t) - m_B \left[-r\omega^2 \left(\cos \omega t + \frac{r}{l} \cos 2\omega t \right) \right] \\ F_{iy} &\equiv -m_A (-r\omega^2 \sin \omega t)\end{aligned}\quad (13.14d)$$

Notice that the x directed inertia forces have primary components at crank frequency and secondary (second harmonic) forces at twice crank frequency. There are also small-magnitude, higher, even harmonics which we truncated in the binomial expansion of the piston displacement function. The force due to the rotating mass at point A has only a primary component.

The **shaking force** was defined in Section 11.8 to be *the sum of all forces acting on the ground plane*. From the free-body diagram for link 1 in Figure 13-12:

$$\begin{aligned}\sum F_{sx} &\equiv -m_A (r\omega^2 \cos \omega t) - m_B \left[r\omega^2 \left(\cos \omega t + \frac{r}{l} \cos 2\omega t \right) \right] \\ \sum F_{sy} &\equiv -m_A (r\omega^2 \sin \omega t) + F_{i41} - F_{i41}\end{aligned}\quad (13.14e)$$

Note that the side force of the piston on the cylinder wall F_{i41} is cancelled by an equal and opposite force passed through the connecting rod and crankshaft to the crank pin. These two forces create a couple that provides the shaking torque. The shaking force $\mathbf{F}_s$ is then just equal to the inertia force.

$$\mathbf{F}_s = \mathbf{F}_i \quad (13.14f)$$

Note that the gas force from equation 13.4 does not contribute to the shaking force. Only inertia forces and external forces are felt as shaking forces. The gas force is an internal force which is cancelled within the mechanism. It acts equally and oppositely on both the piston top and the cylinder head as shown in Figure 13-7 (p. 607).

Program ENGINE calculates the shaking force at constant ω , for any combination of linkage parameters input to it. Figure 13-13 shows the shaking force plot for the same unbalanced built-in example engine as shown in the acceleration plot (Figure 13-8c, p.611). The linkage orientation is the same as in Figure 13-12 (p. 618) with the x axis horizontal. The x component is larger than the y component due to the high acceleration of the piston. The forces are seen to be quite large despite this being a relatively small (0.4 liter per cylinder) engine running at moderate speed (3400 rpm). We will soon investigate techniques to reduce or eliminate this shaking force from the engine. It is an undesirable feature which creates noise and vibration.

13.6 INERTIA AND SHAKING TORQUES

The **inertia torque** results from the action of the inertia forces at a moment arm. The inertia force at point A in Figure 13-12 has two components, radial and tangential. The radial component has no moment arm. The tangential component has a moment arm of crank radius r . If the crank ω is constant, the mass at A will not contribute to inertia torque. The inertia force at B has a nonzero component perpendicular to the cylinder wall except when the piston is at TDC or BDC. As we did for the gas torque, we can express the inertia torque in terms of the couple $-F_{i41}$, F_{i41} whose forces act always perpendicular to the motion of the slider (neglecting friction), and the distance x , which is their instantaneous moment arm. The inertia torque is:

$$\mathbf{T}_{i21} = (F_{i41} \cdot x) \hat{\mathbf{k}} = (-F_{i14} \cdot x) \hat{\mathbf{k}} \quad (13.15a)$$

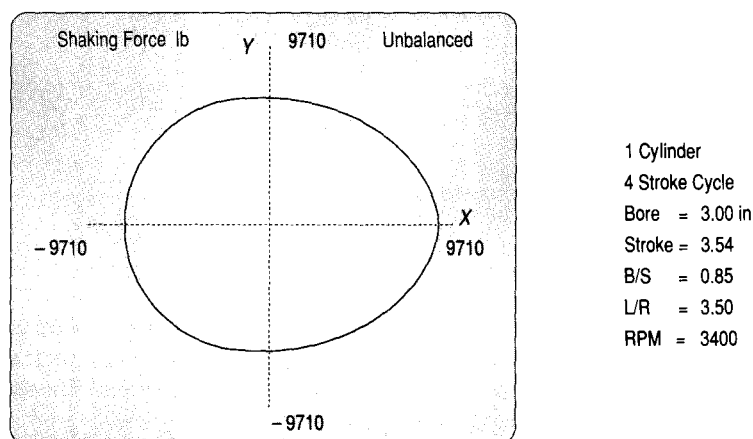


FIGURE 13-13

Shaking force in an unbalanced slider-crank linkage

Substituting for F_{i14} (see Figure 13-12b, p. 618) and for x , (see equation 13.3a, p. 609) we get:

$$\mathbf{T}_{i21} \cong -(-m_B \ddot{x} \tan \phi) \left[l - \frac{r^2}{4l} + r \left(\cos \omega t + \frac{r}{4l} \cos 2\omega t \right) \right] \hat{\mathbf{k}} \quad (13.15b)$$

We previously developed expressions for x *double dot* (equation 13.3e, p. 609) and $\tan \phi$ (equation 13.7d, p. 613) which can now be substituted.

$$\begin{aligned} \mathbf{T}_{i21} \cong m_B \left[-r\omega^2 \left(\cos \omega t + \frac{r}{l} \cos 2\omega t \right) \right] \\ \cdot \left[\frac{r}{l} \sin \omega t \left(1 + \frac{r^2}{2l^2} \sin^2 \omega t \right) \right] \\ \cdot \left[l - \frac{r^2}{4l} + r \left(\cos \omega t + \frac{r}{4l} \cos 2\omega t \right) \right] \hat{\mathbf{k}} \end{aligned} \quad (13.15c)$$

Expanding this and then dropping all terms with coefficients containing r/l to powers higher than one gives the following approximate equation for inertia torque with constant shaft ω :

$$\mathbf{T}_{i21} \cong -m_B r^2 \omega^2 \sin \omega t \left(\frac{r}{2l} + \cos \omega t + \frac{3r}{2l} \cos 2\omega t \right) \hat{\mathbf{k}} \quad (13.15d)$$

This contains products of sine and cosine terms. Putting it entirely in terms of harmonics will be instructive, so substitute the identities:

$$2 \sin \omega t \cos 2\omega t = \sin 3\omega t - \sin \omega t$$

$$2 \sin \omega t \cos \omega t = \sin 2\omega t$$

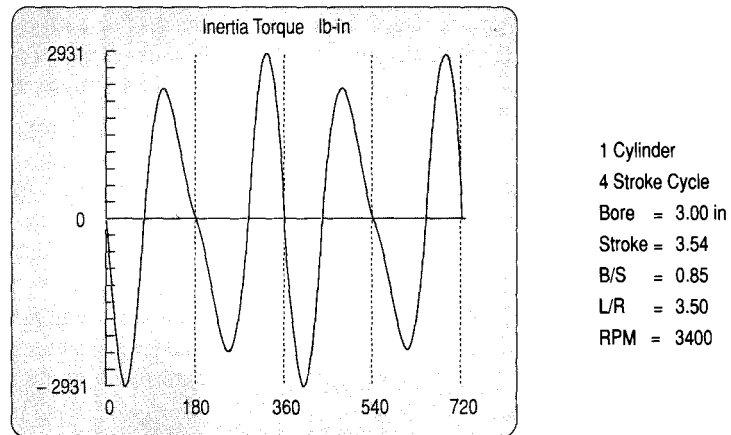
$$\text{to get:} \quad \mathbf{T}_{i21} \cong \frac{1}{2} m_B r^2 \omega^2 \left(\frac{r}{2l} \sin \omega t - \sin 2\omega t - \frac{3r}{2l} \sin 3\omega t \right) \hat{\mathbf{k}} \quad (13.15e)$$

This shows that the **inertia torque** has a third harmonic term as well as a first and second. The second harmonic is the dominant term as it has the largest coefficient because r/l is always less than $2/3$.

The **shaking torque** is equal to the inertia torque.

$$\mathbf{T}_s = \mathbf{T}_{i21} \quad (13.15f)$$

Program ENGINE calculates the inertia torque from equation B.15e. Figure 13-14 shows a plot of the inertia torque for this built-in example engine. Note the dominance of the second harmonic. The ideal magnitude for the inertia torque is zero, as it is parasitic. Its average value is always zero, so it *contributes nothing to the net driving torque*. It merely creates large positive and negative oscillations in the total torque which increase vibration and roughness. We will soon investigate means to reduce or eliminate this inertia and shaking torque in our engine designs. It is possible to cancel their effects by proper arrangement of the cylinders in a multicylinder engine, as will be explored in the next chapter.

**FIGURE 13-14**

Inertia torque in the slider-crank linkage

13.7 TOTAL ENGINE TORQUE

The total engine torque is the sum of the gas torque and the inertia torque.

$$T_{total} = T_g + T_i \quad (13.16)$$

The gas torque is less sensitive to engine speed than is the inertia torque, which is a function of ω^2 . So the relative contributions of these two components to the total torque will vary with engine speed. Figure 13-15a shows the total torque for this example engine plotted by program ENGINE for an idle speed of 800 rpm. Compare this to the gas torque plot of the same engine in Figure 13-9a (p. 612). The inertia torque component is negligible at this slow speed compared to the gas torque component. Figure 13-15c shows the same engine run at 6000 rpm. Compare this to the plot of inertia torque in Figure 13-14. The inertia torque component is dominating at this high speed. At the midrange speed of 3400 rpm (Figure 13-15b), a mix of both components is seen.

13.8 FLYWHEELS

We saw in Section 11.11 (p. 548) that large oscillations in the torque-time function can be significantly reduced by the addition of a flywheel to the system. The single-cylinder engine is a prime candidate for the use of a flywheel. The intermittent nature of its power strokes makes one mandatory as it will store the kinetic energy needed to carry the piston through the Otto cycle's exhaust, intake, and compression strokes during which work must be done on the system. Even the two-stroke engine needs a flywheel to drive the piston up on the compression stroke.

The procedure for designing an engine flywheel is identical to that described in Section 11.11 for the fourbar linkage. The total torque function for one revolution of the crank is integrated, pulse by pulse, with respect to its average value. These integrals represent energy fluctuations in the system. The maximum change in energy under the torque curve during one cycle is the amount needed to be stored in the flywheel. Equa-

tion 11.20c expresses this relationship. Program ENGINE does the numerical integration of the total torque function and presents a table similar to the one shown in Figure 11-11 (p. 551). These data and the designer's choice of a coefficient of fluctuation k (see equation 11.19) are all that are needed to solve equations 11.20 and 11.21 for the required moment of inertia of the flywheel.

The calculation must be done at some average crank θ . Since the typical engine operates over a wide range of speeds, some thought needs to be given to the most appropriate speed to use in the flywheel calculation. The flywheel's stored kinetic energy is proportional to ω^2 (see equation 11.17). Thus at high speeds a flywheel can have a small moment of inertia and still be effective. The slowest operating speed will require the largest flywheel and should be the one used in the computation of required flywheel size.

Program ENGINE plots the flywheel-smoothed total torque for a user-supplied coefficient of fluctuation k . Figure 13-16 shows the smoothed torque functions for $k = 0.05$ corresponding to the unsmoothed ones in Figure 13-15. Note that the smoothed curves shown for each engine speed are what would result with the flywheel size necessary to obtain that coefficient of fluctuation at that speed. In other words, the flywheel applied to the 800-rpm engine is much larger than the one on the 6000-rpm engine, in these plots. Compare corresponding rows (speeds) between Figures 13-15 and 13-16 to see the effect of the addition of a flywheel. But do not directly compare parts a, b, and c within Figure 13-16 as to the amount of smoothing since the flywheel sizes used are different at each operating speed.

An engine flywheel is usually designed as a flat disk, bolted to one end of the crankshaft. One flywheel face is typically used for the clutch to run against. The clutch is a friction device which allows disconnection of the engine from the drive train (the wheels of a vehicle) when no output is desired. The engine can then remain running at idle speed with the vehicle or output device stopped. When the clutch is engaged, all engine torque is transmitted through it, by friction, to the output shaft.

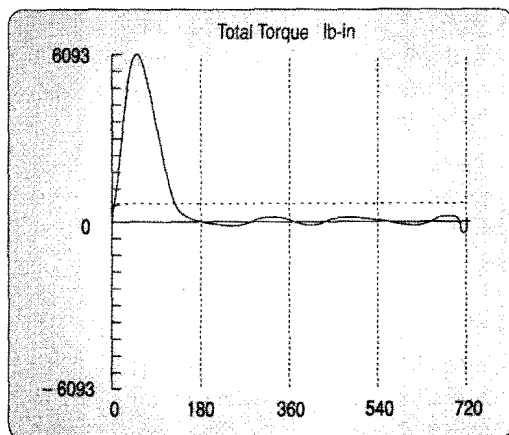
13.9 PIN FORCES IN THE SINGLE-CYLINDER ENGINE

In addition to calculating the overall effects on the ground plane of the dynamic forces present in the engine, we also need to know the magnitudes of the forces at the pin joints. These forces will dictate the design of the pins and the bearings at the joints. Though we were able to lump the mass due to both conrod and piston, or conrod and crank, at points A and B for an overall analysis of the linkage's effects on the ground plane, we cannot do so for the pin force calculations. This is because the pins feel the effect of the conrod pulling on one "side" and the piston (or crank) pulling on the other "side" of the pin as shown in Figure 13-17. Thus we must separate the effects of the masses of the links joined by the pins.

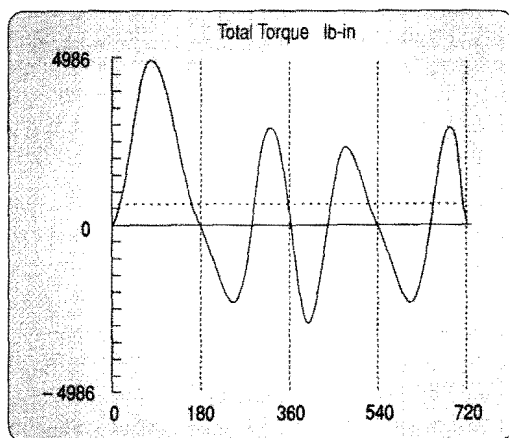
We will calculate the effect of each component due to the various masses, and the gas force and then superpose them to obtain the complete pin force at each joint. We need a bookkeeping system to keep track of all these components. We have already used some subscripts for these forces, so we will retain them and add others. The resultant bearing loads have the following components:

- 1 The gas force components, with the subscript g , as in F_g .
- 2 The inertia force due to the piston mass, with subscript ip , as in F_{ip} .

(a) 800 rpm

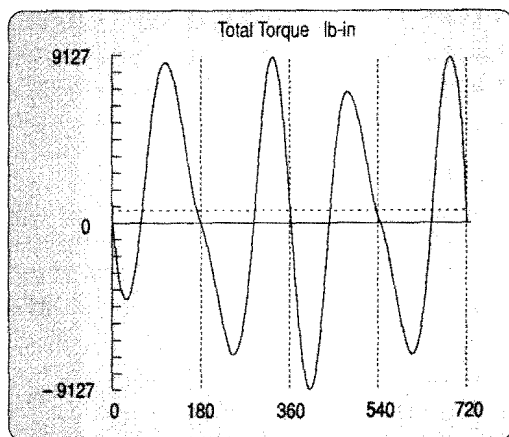


(b) 3400 rpm



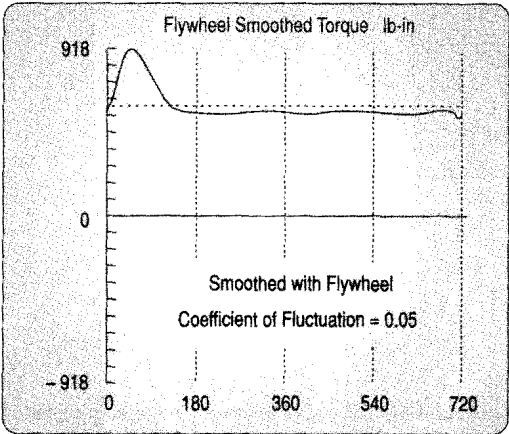
1 Cylinder
 4 Stroke Cycle
 Bore = 3.00 in
 Stroke = 3.54
 B/S = 0.85
 L/R = 3.50

(c) 6000 rpm

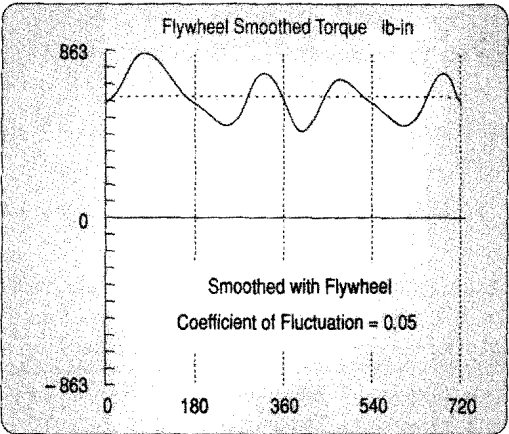
**FIGURE 13-15**

The total torque function's shape and magnitude vary with crankshaft speed

(a) 800 rpm



(b) 3400 rpm



1 Cylinder
4 Stroke Cycle
Bore = 3.00 in
Stroke = 3.54
B/S = 0.85
L/R = 3.50

(c) 6000 rpm

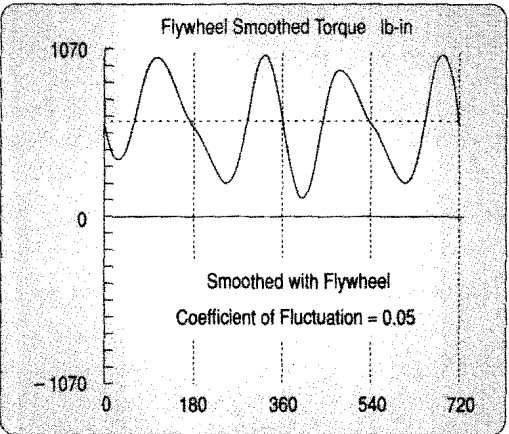
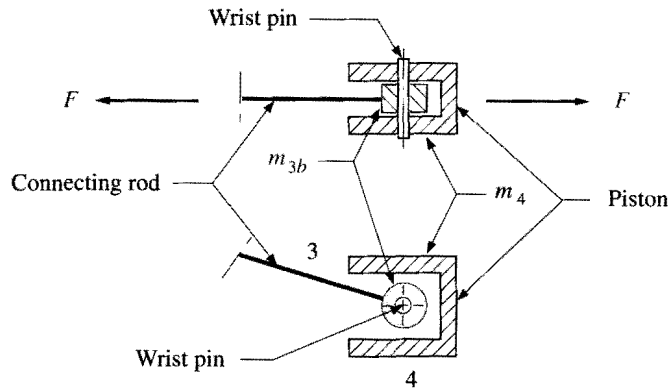


FIGURE 13-16

The total torque function's shape and magnitude vary with crankshaft speed

**FIGURE 13-17**

Forces on a pivot pin

- 3 The inertia force due to the mass of the conrod at the wrist pin, with subscript *iw*, as in F_{iw} .
- 4 The inertia force due to the mass of the conrod at the crank pin, with subscript *ic*, as in F_{ic} .
- 5 The inertia force due to the mass of the crank at the crank pin, with subscript *ir*, as in F_{ir} .

The link number designations will be added to each subscript in the same manner as before, indicating the link from which the force is coming as the first number and the link being analyzed as the second. (See Section 11.2 for further discussion of this notation.)

Figure 13-18 shows the free-body diagrams for the forces due only to the acceleration of the mass of the piston, m_4 . Those components are:

$$\mathbf{F}_{ip41} = -m_4 a_B \tan \phi \hat{\mathbf{j}} \quad (13.17a)$$

$$\mathbf{F}_{ip34} = m_4 a_B \hat{\mathbf{i}} - m_4 a_B \tan \phi \hat{\mathbf{j}} \quad (13.17b)$$

$$\mathbf{F}_{ip32} = -\mathbf{F}_{ip34} \quad (13.17c)$$

$$\mathbf{F}_{ip21} = \mathbf{F}_{ip32} \quad (13.17d)$$

Figure 13-19 shows the free-body diagrams for the forces due only to the acceleration of the mass of the conrod located at the wrist pin, m_{3b} . Those components are:

$$\mathbf{F}_{iw41} = -m_{3b} a_B \tan \phi \hat{\mathbf{j}} \quad (13.18a)$$

$$\mathbf{F}_{iw34} = \mathbf{F}_{iw41} \quad (13.18b)$$

$$\mathbf{F}_{iw32} = -m_{3b} a_B \hat{\mathbf{i}} + m_{3b} a_B \tan \phi \hat{\mathbf{j}} \quad (13.18c)$$

$$\mathbf{F}_{iw21} = \mathbf{F}_{iw32} \quad (13.18d)$$

Figure 13-20a shows the free-body diagrams for the forces due only to the acceleration of the mass of the conrod located at the crankpin, m_{3a} . That component is:

$$\mathbf{F}_{ic32} = -m_{3a} \mathbf{a}_A \quad (13.19a)$$

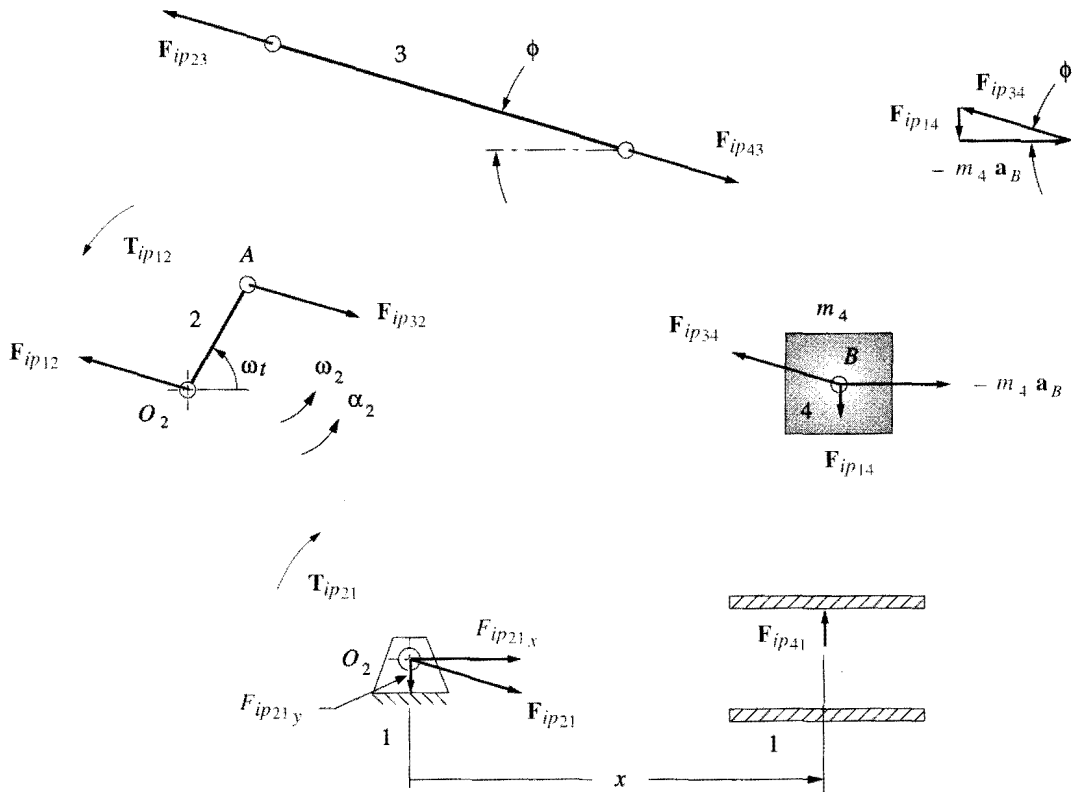


FIGURE 13-18

Free-body diagrams for forces due to the piston mass

Substitute equation 13.13 (p. 619) for $\mathbf{a}_A$:

$$\mathbf{F}_{ic32} = m_{3a} r \omega^2 (\cos \omega t \hat{\mathbf{i}} + \sin \omega t \hat{\mathbf{j}}) \quad (13.19b)$$

Figure 13-20b shows the free-body diagrams for the forces due only to the acceleration of the lumped mass of the crank at the crankpin, m_{2a} . These affect only the main pin at O_2 . That component is:

$$\mathbf{F}_{ir21} = m_{2a} r \omega^2 (\cos \omega t \hat{\mathbf{i}} + \sin \omega t \hat{\mathbf{j}}) \quad (13.19c)$$

The gas force components were shown in Figure 13-7 (p. 607) and defined in equations 13.5 (p. 611).

We can now sum the components of the forces at each pin joint. For the sidewall force $\mathbf{F}_{41}$ of the piston against the cylinder wall:

$$\begin{aligned} \mathbf{F}_{41} &= \mathbf{F}_{g41} + \mathbf{F}_{ip41} + \mathbf{F}_{iw41} \\ &= -F_g \tan \phi \hat{\mathbf{j}} - m_4 a_B \tan \phi \hat{\mathbf{j}} - m_{3b} a_B \tan \phi \hat{\mathbf{j}} \\ &= -[(m_4 + m_{3b}) a_B + F_g] \tan \phi \hat{\mathbf{j}} \end{aligned} \quad (13.20)$$

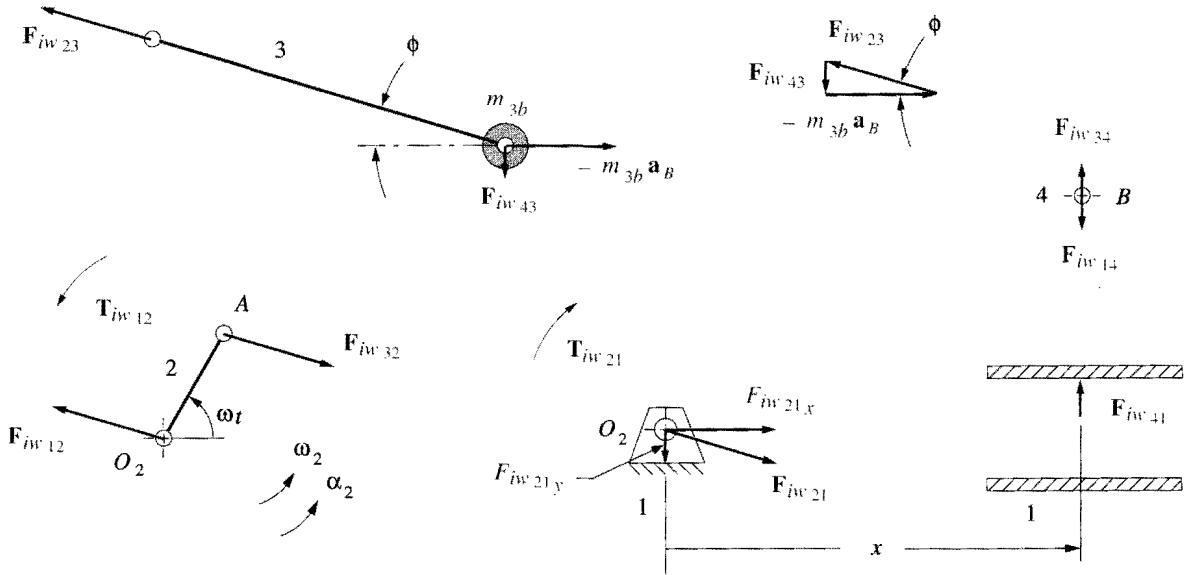


FIGURE 13-19

Free-body diagrams for forces due to the conrod mass concentrated at the wrist pin

The total force $\mathbf{F}_{34}$ on the wrist pin is:

$$\begin{aligned}
 \mathbf{F}_{34} &= \mathbf{F}_{g34} + \mathbf{F}_{ip34} + \mathbf{F}_{iw34} \\
 &= (F_g \hat{\mathbf{i}} - F_g \tan \phi \hat{\mathbf{j}}) + (m_4 a_B \hat{\mathbf{i}} - m_4 a_B \tan \phi \hat{\mathbf{j}}) + (-m_{3b} a_B \tan \phi \hat{\mathbf{j}}) \\
 &= (F_g + m_4 a_B) \hat{\mathbf{i}} - [F_g + (m_4 + m_{3b}) a_B] \tan \phi \hat{\mathbf{j}}
 \end{aligned} \quad (13.21)$$

The total force $\mathbf{F}_{32}$ on the crankpin is:

$$\begin{aligned}
 \mathbf{F}_{32} &= \mathbf{F}_{g32} + \mathbf{F}_{ip32} + \mathbf{F}_{iw32} + \mathbf{F}_{ic32} \\
 &= (-F_g \hat{\mathbf{i}} + F_g \tan \phi \hat{\mathbf{j}}) + (-m_4 a_B \hat{\mathbf{i}} + m_4 a_B \tan \phi \hat{\mathbf{j}}) \\
 &\quad + (-m_{3b} a_B \hat{\mathbf{i}} + m_{3b} a_B \tan \phi \hat{\mathbf{j}}) + [m_{3a} r \omega^2 (\cos \omega t \hat{\mathbf{i}} + \sin \omega t \hat{\mathbf{j}})] \\
 &= [m_{3a} r \omega^2 \cos \omega t - (m_{3b} + m_4) a_B - F_g] \hat{\mathbf{i}} \\
 &\quad + \{m_{3a} r \omega^2 \sin \omega t + [(m_{3b} + m_4) a_B + F_g] \tan \phi\} \hat{\mathbf{j}}
 \end{aligned} \quad (13.22)$$

The total force $\mathbf{F}_{21}$ on the main pin is:

$$\mathbf{F}_{21} = \mathbf{F}_{32} + \mathbf{F}_{ir21} \quad (13.23)$$

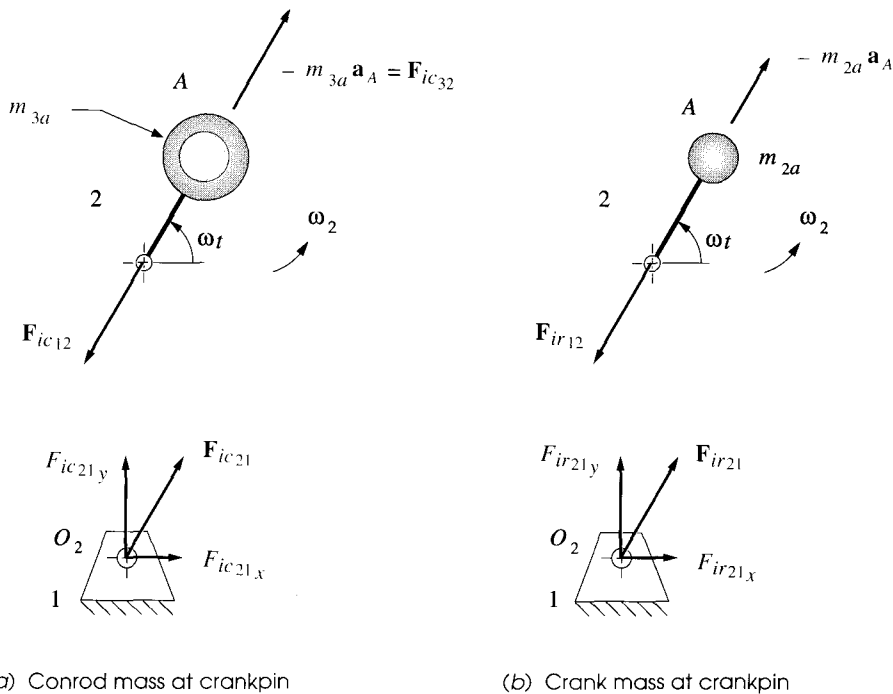


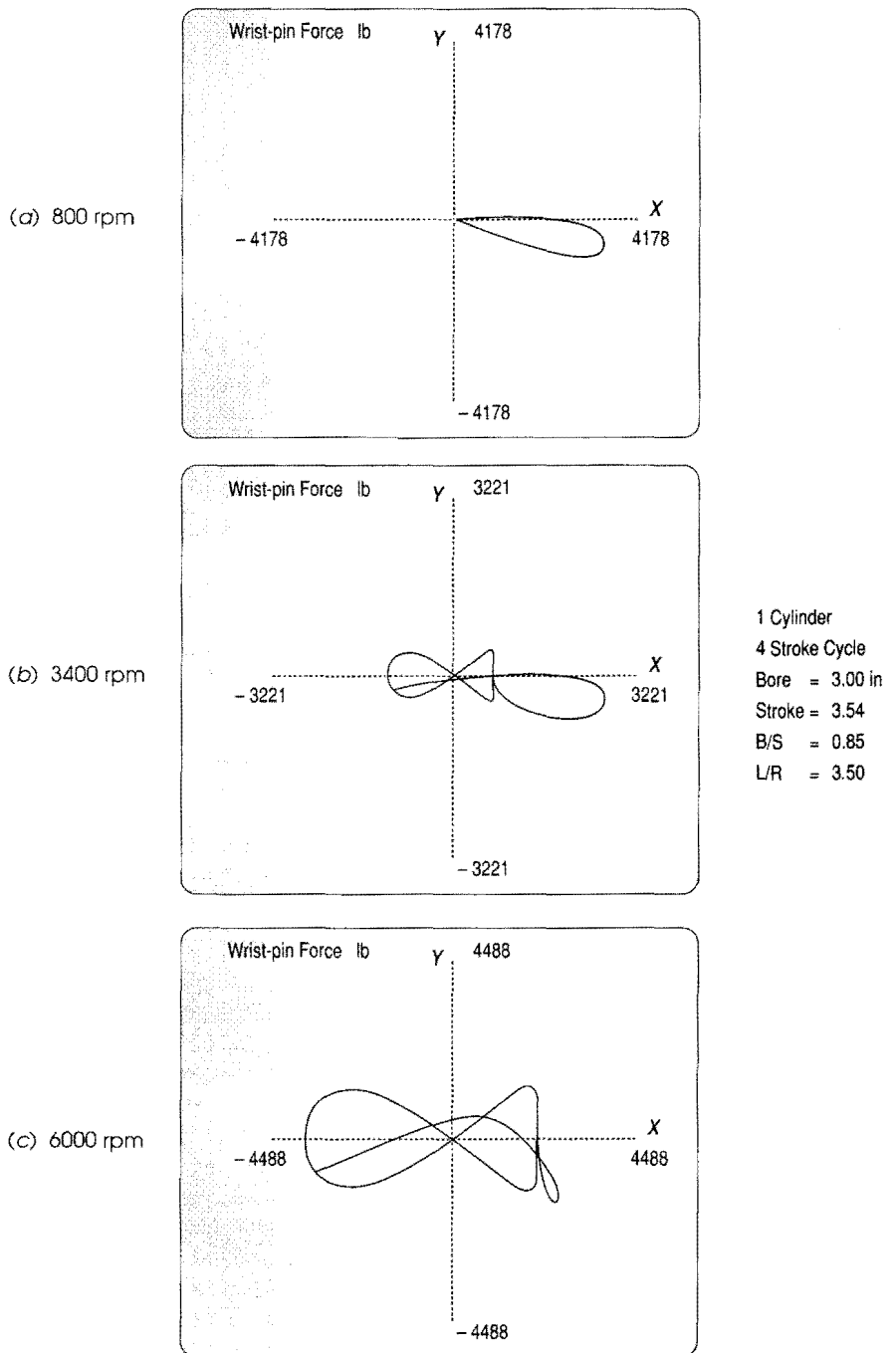
FIGURE 13-20

Free-body diagrams for forces due to masses at the crankpin

Note that, unlike the inertia force in equation 13.14 (p. 619), which was unaffected by the gas force, these pin forces **are** a function of the gas force as well as of the $-ma$ forces. Engines with larger piston diameters will experience greater pin forces as a result of the explosion pressure acting on their larger piston area.

Program ENGINE calculates the pin forces on all joints using equations 13.20 to 13.23. Figure 13-21 shows the wrist-pin force on the same unbalanced engine example as shown in previous figures, for three engine speeds. The "bow tie" loop is the inertia force and the "teardrop" loop is the gas force portion of the force curve. An interesting trade-off occurs between the gas force components and the inertia force components of the pin forces. At a low speed of 800 rpm (Figure 13-21a), the gas force dominates as the inertia forces are negligible at small ϕ . The peak wrist-pin force is then about 4200 lb. At high speed (6000 rpm), the inertia components dominate and the peak force is about 4500 lb (Figure 13-21c). But at a midrange speed (3400 rpm), the inertia force cancels some of the gas force and the peak force is only about 3200 lb (Figure 13-21b). These plots show that the pin forces can be quite large even in a moderately sized (0.4 liter/cylinder) engine. The pins, links, and bearings all have to be designed to withstand hundreds of millions of cycles of these reversing forces without failure.

Figure 13-22 shows further evidence of the interaction of the gas forces and inertia forces on the crankpin and the wrist pin. Figures 13-22a and 13-22c show the variation in the inertia force component on the crankpin and wrist pin, respectively, over one full

**FIGURE 13-21**

Forces on the wrist pin of the single-cylinder engine at various speeds

revolution of the crank as the engine speed is increased from idle to redline. Figure 13-22b and d show the variation in the total force on the same respective pins with both the inertia and gas force components included. These two plots show only the first 90° of crank revolution where the gas force in a four-stroke cylinder occurs. Note that the gas force and inertia force components counteract one another resulting in one particular speed where the total pin force is a minimum during the power stroke. This is the same phenomenon as seen in Figure 13-21.

13.10 BALANCING THE SINGLE-CYLINDER ENGINE

The derivations and figures in the preceding sections have shown that significant forces are developed both on the pivot pins and on the ground plane due to the gas forces and the inertia and shaking forces. Balancing will not have any effect on the gas forces, which are internal, but it can have a dramatic effect on the inertia and shaking forces. The main pin force can be reduced, but the crankpin and wrist pin forces will be unaffected by any crankshaft balancing done. Figure 13-13 (p. 620) shows the unbalanced shaking force as felt on the ground plane of our OA-liter-per-cylinder example engine from program ENGINE. It is about 9700 lb even at the moderate speed of 3400 rpm. At 6000 rpm it increases to over 30 000 lb. The methods of Chapter 12 can be applied to this mechanism to balance the members in pure rotation and reduce these large shaking forces.

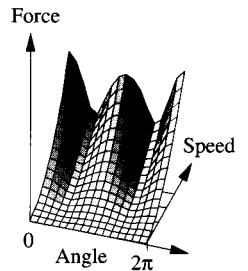
Figure 13-23a shows the dynamic model of our slider-crank with the conrod mass lumped at both crankpin A and wrist pin B. We can consider this single-cylinder engine to be a single-plane device, thus suitable for static balancing (see Section 13.1). It is straightforward to statically balance the crank. We need a balance mass at some radius, 180° from the lumped mass at point A whose mr product is equal to the product of the mass at A and its radius r . Applying equation 13.2 to this simple problem we get:

$$m_{bal}r_{bal} = -m_A r_A \tag{13.24}$$

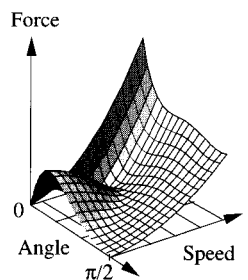
Any combination of mass and radius which gives this product, placed at 180° from point A will balance the crank. For simplicity of example, we will use a balance radius equal to r . Then a mass equal to m_A placed at A' will exactly balance the rotating masses. The CG of the crank will then be at the fixed pivot O2 as shown in Figure 13-23a. In a real crankshaft, actually placing the counterweight at this large a radius would not work. The balance mass has to be kept close to the centerline to clear the piston at BDC. Figure 13-2c shows the shape of typical crankshaft counterweights.

Figure 13-24a shows the shaking force from the same engine as in Figure 13-13 after the crank has been exactly balanced in this manner. The Y component of the shaking force has been reduced to zero and the x component to about 3300 lb at 3400 rpm. This is a factor of three reduction over the unbalanced engine. Note that the only source of Y directed inertia force is the rotating mass at point A of Figure 13-23 (see equations 13.14, p. 619). What remains after balancing the rotating mass is the force due to the acceleration of the piston and conrod masses at point B of Figure 13-23 which are in linear translation along the X axis, as shown by the inertia force $-m_B a_B$ at point B in that figure.

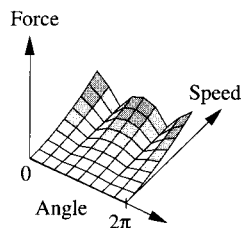
To completely eliminate this reciprocating unbalanced shaking force would require the introduction of another reciprocating mass, which oscillates 180° out of phase with the piston. Adding a second piston and cylinder, properly arranged, can accomplish this.



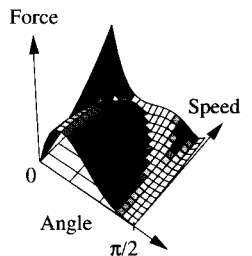
(a) Crankpin inertia force



(b) Crankpin total force



(c) Wrist-pin inertia force



(d) Wrist-pin total force

FIGURE 13-22
Pin force variation

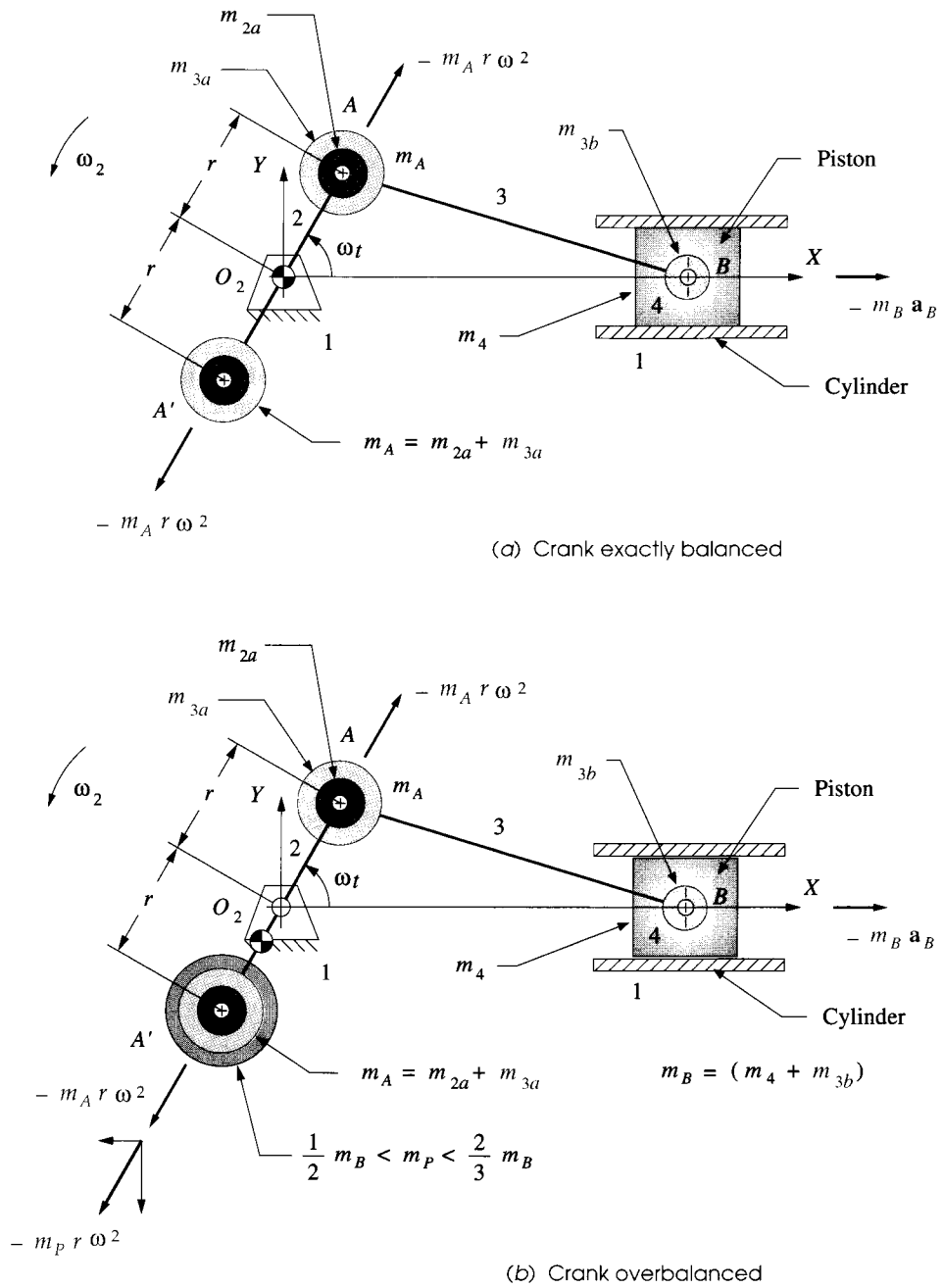


FIGURE 13-23

Balancing and overbalancing the single-cylinder engine

One of the principal advantages of multicylinder engines is their ability to reduce or eliminate the shaking forces. We will investigate this in the next chapter.

In the single-cylinder engine, there is no way to completely eliminate the reciprocating unbalance with a single, rotating counterweight, but we can reduce the shaking force still further. Figure 13-23b shows an additional amount of mass m_P added to the counterweight at point A' . (Note that the crank's CG has now moved away from the fixed pivot.) This extra balance mass creates an additional inertia force ($-m_P r \omega^2$) as is shown, broken into x and y components, in the figure. The y component is not opposed by any other inertia forces present, but the x component will always be opposite to the reciprocating inertia force at point B . Thus this extra mass, m_P , which *overbalances the crank*, will reduce the x directed shaking force at the expense of adding back some y directed shaking force. This is a useful trade-off as the direction of the shaking force is usually of less concern than is its magnitude. Shaking forces create vibrations in the supporting structure which are transmitted through it and modified by it. As an example, it is unlikely that you could define the direction of a motorcycle engine's shaking forces by feeling their resultant vibrations in the handlebars. But you **will** detect an increase in the magnitude of the shaking forces from the larger amplitude of vibration they cause in the cycle frame.

The correct amount of additional “overbalance” mass needed to minimize the peak shaking force, regardless of its direction, will vary with the particular engine design. It will usually be between one-half and two-thirds of the reciprocating mass at point B (piston plus conrod at wrist pin), if placed at the crank radius r . Of course, once this mass-radius product is determined, it can be achieved with any combination of mass and radius. Figure 13-24b shows the minimum shaking force achieved for this engine with the addition of 65.5% of the mass at B acting at radius r . The shaking force has now been reduced to 1700 lb at 3400 rpm, which is **18% of its original unbalanced value**. The

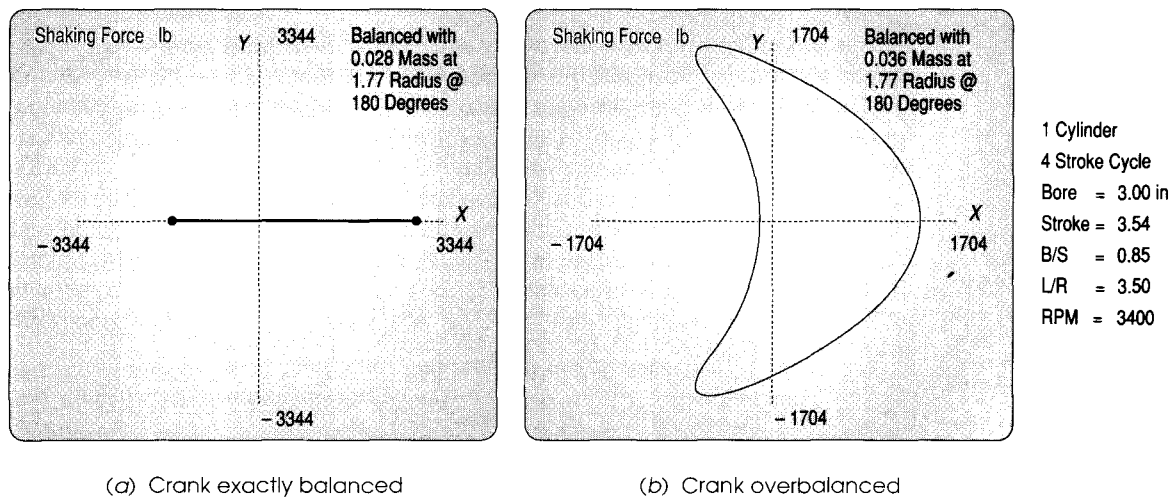


FIGURE 13-24
Effects of balancing and overbalancing on the shaking force in the slider-crank linkage

benefits of balancing, and of overbalancing in the case of the single-cylinder engine, should now be obvious.

13.11 DESIGN TRADE-OFFS AND RATIOS

In the design of any system or device, no matter how simple, there will always be conflicting demands, requirements, or desires which must be traded off to achieve the best design compromise. This single-cylinder engine is no exception. There are two dimensionless design ratios which can be used to characterize an engine's dynamic behavior in a general way. The first is the crank/conrod ratio r/l , introduced in Section 13.2 (p. 605), or its inverse, the **conrod/crank ratio** l/r . The second is the **bore/stroke ratio** B/S .

Conrod/Crank Ratio

The crank/conrod ratio r/l appears in all the equations for acceleration, forces, and torques. In general, the smaller the r/l ratio, the smoother will be the acceleration function and thus all other factors which it influences. Program ENGINE uses the inverse of this ratio as an input parameter. The **conrod/crank ratio** l/r must be greater than about two to obtain acceptable transmission angles in the slider-crank linkage. The ideal value for l/r from a kinematic standpoint would be infinity as that would result in the piston acceleration function being a pure harmonic. The second and all subsequent harmonic terms in equations 13.3 (p. 619) would be zero in this event, and the peak value of acceleration would be a minimum. However, an engine this tall would not package very well, and package considerations often dictate the maximum value of the l/r ratio. Most engines will have an l/r ratio between three and five which values give acceptable smoothness in a reasonably short engine.

Bore/Stroke Ratio

The bore B of the cylinder is essentially equal to the diameter of the piston. (There is a small clearance.) The stroke S is defined as the distance travelled by the piston from TDC to BDC and is twice the crank radius, $S = 2r$. The bore appears in the equation for gas force (equation 13.4, p. 610) and thus also affects gas torque. The crank radius appears in every equation. An engine with a B/S ratio of 1 is referred to as a "square" engine. If B/S is larger than 1, it is "oversquare"; if less, "undersquare." The designer's choice of this ratio can have a significant effect on the dynamic behavior of the engine. Assuming that the displacement, or stroke volume V , of the engine has been chosen and is to remain constant, this displacement can be achieved with an infinity of combinations of bore and stroke ranging from a "pancake" piston with tiny stroke to a "pencil" piston with very long stroke.

$$V = \frac{\pi B^2}{4} S \quad (13.25)$$

There is a classic design trade-off here between B and S for a constant stroke volume V . A large bore and small stroke will result in high gas forces which will affect pin forces adversely. A large stroke and small bore will result in high inertia forces which will affect pin forces (as well as other forces and torques) adversely. So there should be

an optimum value for the B/S ratio in each case, which will minimize these adverse effects. Most production engines have B/S ratios in the range of about 0.75 to 1.5.

It is left as an exercise for the student to investigate the effects of variation in the B/S and l/r ratios on forces and torques in the system. Program ENGINE will demonstrate the effects of changes made independently to each of these ratios, while all other design parameters are held constant. The student is encouraged to experiment with the program to gain insight into the role of these ratios in the dynamic performance of the engine.

Materials

There will always be a strength/weight trade-off. The forces in this device can be quite high, due both to the explosion and to the inertia of the moving elements. We would like to keep the masses of the parts as low as possible as the accelerations are typically very high, as seen in Figure 13-8c (p. 611). But the parts must be strong enough to withstand the forces, so materials with good strength-to-weight ratios are needed. Pistons are usually made of an aluminum alloy, either cast or forged. Conrods are most often cast iron or forged steel, except in very small engines (lawn mower, chain saw, motorcycle) where they may be aluminum alloy. High-performance engines may have titanium connecting rods. Crankshafts are usually cast iron or forged steel, and wrist pins are of hardened steel tubing or rod. Plain bearings of a special soft, nonferrous metal alloy called babbitt are usually used. In the four-stroke engine these are pressure lubricated with oil pumped through drilled passageways in the block, crankshaft, and connecting rods. In the two-stroke engine, the fuel carries the lubricant to these parts. Engine blocks are cast iron or cast aluminum alloy. The chrome-plated steel piston rings seal and wear well against cast iron cylinders. Most aluminum blocks are fitted with cast iron liners around the cylinder bores. Some are unlined and made of a high-silicon aluminum alloy which is specially cooled after casting to precipitate the hard silicon in the cylinder walls for wear resistance.

13.12 BIBLIOGRAPHY

- 1 Heywood, J. B. (1988). *Internal Combustion Engine Fundamentals*. McGraw-Hill: New York.
- 2 Taylor, C. F. (1966). *The Internal Combustion Engine in Theory and Practice*. MIT Press: Cambridge, MA.

13.13 PROBLEMS

- *†13-1 A slider-crank linkage has $r = 3$ and $l = 12$. It has an angular velocity of 200 rad/sec at time $t = 0$. Its initial crank angle is zero. Calculate the piston acceleration at $t = 1$ sec. Use two methods, the exact solution, and the approximate Fourier series solution and compare the results.
- †13-2 Repeat Problem 13-1 for $r = 4$ and $l = 15$ and $t = 0.9$ sec.
- *†13-3 A slider-crank linkage has $r = 3$ and $l = 12$, and a piston bore $B = 2$. The peak gas pressure in the cylinder occurs at a crank angle of 10° and is 1000 pressure units. Calculate the gas force and gas torque at this position.

* Answers in Appendix F.

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs.

- †13-4 A slider-crank linkage has $r = 4$ and $l = 15$, and a piston bore $B = 3$. The peak gas pressure in the cylinder occurs at a crank angle of 5° and is 600 pressure units. Calculate the gas force and gas torque at this position.
- *†13-5 Repeat Problem 13-3 using the exact method of calculation of gas torque and compare its result to that obtained by the approximate expression in equation 13.8b (p. 613). What is the percent error?
- †13-6 Repeat Problem 13-4 using the exact method of calculation of gas torque and compare its result to that obtained by the approximate expression in equation 13.8b (p. 613). What is the percent error?
- *†13-7 A connecting rod of length $l = 12$ has a mass $m_3 = 0.020$. Its mass moment of inertia is 0.620. Its CG is located at $0.4\ l$ from the crankpin, point A.
- Calculate an exact dynamic model using two lumped masses, one at the wrist pin, point B, and one at whatever other point is required. Define the lumped masses and their locations.
 - Calculate an approximate dynamic model using two lumped masses, one at the wrist pin, point B, and one at the crankpin, point A. Define the lumped masses and their locations.
 - Calculate the error in the mass moment of inertia of the approximate model as a percentage of the original mass moment of inertia.
- †13-8 Repeat Problem 13-7 for these data: $l = 15$, mass $m_3 = 0.025$, mass moment of inertia is 1.020. Its CG is located at $0.25\ l$ from the crankpin, point A.
- *†13-9 A crank of length $r = 3.5$ has a mass $m_2 = 0.060$. Its mass moment of inertia about its pivot is 0.300. Its CG is located at $0.30\ r$ from the main pin, point O_2 . Calculate a statically equivalent two-lumped mass dynamic model with the lumps placed at the main pin and crankpin. What is the percent error in the model's moment of inertia about the crank pivot.
- †13-10 Repeat Problem 13-9 for a crank length $r = 4$, a mass $m_2 = 0.050$, a mass moment of inertia about its pivot of 0.400. Its CG is located at $0.40\ r$ from the main pin, point O_2 .
- *†13-11 Combine the data from problems 13-7 and 13-9. Run the linkage at a constant 2000 rpm. Calculate the inertia force and inertia torque at $\omega t = 45^\circ$. Piston mass = 0.012.
- †13-12 Combine the data from problems 13-7 and 13-10. Run the linkage at a constant 3000 rpm. Calculate the inertia force and inertia torque at $\omega t = 30^\circ$. Piston mass = 0.019.
- †13-13 Combine the data from problems 13-8 and 13-9. Run the linkage at a constant 2500 rpm. Calculate the inertia force and inertia torque at $\omega t = 24^\circ$. Piston mass = 0.023.
- *†13-14 Combine the data from problems 13-8 and 13-10. Run the linkage at a constant 4000 rpm. Calculate the inertia force and inertia torque at $\omega t = 18^\circ$. Piston mass = 0.015.
- †13-15 Combine the data from problems 13-7 and 13-9. Run the linkage at a constant 2000 rpm. Calculate the pin forces at $\omega t = 45^\circ$. Piston mass = 0.022. $F_g = 300$.
- †13-16 Combine the data from problems 13-7 and 13-10. Run the linkage at a constant 3000 rpm. Calculate the pin forces at $\omega t = 30^\circ$. Piston mass = 0.019. $F_g = 600$.

* Answers in Appendix F.

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs.

†13-17 Combine the data from problems 13-8 and 13-9. Run the linkage at a constant 2500 rpm. Calculate the pin forces at $\omega t = 24^\circ$. Piston mass = 0.032. $F_g = 900$.

†13-18 Combine the data from problems 13-8 and 13-10. Run the linkage at a constant 4000 rpm. Calculate the pin forces at $\omega t = 18^\circ$. Piston mass = 0.014. $F_g = 1200$.

*†‡13-19 Using the data from Problem 13-11:

- Exactly balance the crank and recalculate the inertia force.
- Overbalance the crank with approximately two-thirds of the mass at the wrist pin placed at radius $-r$ on the crank and recalculate the inertia force.
- Compare these results to those for the unbalanced crank.

†‡13-20 Repeat Problem 13-19 using the data from Problem 13-12.

†‡13-21 Repeat Problem 13-19 using the data from Problem 13-13.

†‡13-22 Repeat Problem 13-19 using the data from Problem 13-14.

†13-23 Combine the necessary equations to develop expressions that show how each of these dynamic parameters varies as a function of the crank/conrod ratio alone:

- Piston acceleration
- Inertia force
- Inertia torque
- Pin forces

Plot the functions. Check your conclusions with program ENGINE.

Hint: Consider all other parameters to be temporarily constant. Set the crank angle to some value such that the gas force is nonzero.

†13-24 Combine the necessary equations to develop expressions that show how each of these dynamic parameters varies as a function of the bore/stroke ratio alone:

- Gas force
- Gas torque
- Inertia force
- Inertia torque
- Pin forces

Plot the functions. Check your conclusions with program ENGINE.

Hint: Consider all other parameters to be temporarily constant. Set the crank angle to some value such that the gas force is nonzero.

†13-25 Develop an expression to determine the optimum bore/stroke ratio to minimize the wrist pin force. Plot the function.

†‡13-26 Use program ENGINE, your own computer program, or an equation solver to calculate the maximum value and the polar-plot shape of the force on the main pin of a 1-in³ displacement, single-cylinder engine with bore = 1.12838 in for the following situations.

- Piston, conrod and crank masses = 0
- Piston mass = 1 blob, conrod and crank masses = 0
- Conrod mass = 1 blob, piston and crank masses = 0
- Crank mass = 1 blob, conrod and piston masses = 0

* Answers in Appendix F.

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs.

‡ These problems are suited to solution using program ENGINE which is on the attached CD-ROM.

Place the *CG* of the crank at $0.5 r$ and the conrod at $0.33 l$. Compare and explain the differences in the main pin force under these different conditions with reference to the governing equations.

†13-27 Repeat Problem 13-26 for the crankpin.

†13-28 Repeat Problem 13-26 for the wrist pin.

†‡13-29 Use program ENGINE, your own computer program, or an equation solver to calculate the maximum value and the polar plot shape of the force on the main pin of a 1 in^3 single-cylinder engine with bore = 1.12838 in for the following situations.

- Engine unbalanced.
- Crank exactly balanced against mass at crankpin.
- Crank optimally overbalanced against masses at crankpin and wrist pin.

Piston, conrod, and crank masses = 1. Place the *CG* of the crank at $0.5 r$ and the conrod at $0.33 l$. Compare and explain the differences in the main pin force under these conditions with reference to the governing equations.

†‡13-30 Repeat Problem 13-29 for the crankpin force.

†‡13-31 Repeat Problem 13-29 for the wrist pin force.

†‡13-32 Repeat Problem 13-29 for the shaking force.

13.14 PROJECTS

These are loosely structured design problems intended for solution using program ENGINE. All involve the design of a single-cylinder engine and differ only in the specific data for the engine. The general problem statement is:

Design a single-cylinder engine for a specified displacement and stroke cycle. Optimize the conrod/crank ratio and bore/stroke ratio to minimize shaking forces, shaking torque, and pin forces, also considering package size. Design your link shapes and calculate realistic dynamic parameters (mass, CG location, moment of inertia) for those links using the methods shown in Chapter 11 and Section 12.12. Dynamically model the links as described in this chapter. Balance or overbalance the linkage as needed to achieve these results. Overall smoothness of total torque is desired. Design and size a minimum weight flywheel by the method of Chapter 12 to smooth the total torque. Write an engineering report on your design.

P13-1 Two-stroke cycle with a displacement of 0.125 liters.

P13-2 Four-stroke cycle with a displacement of 0.125 liters.

P13-3 Two-stroke cycle with a displacement of 0.25 liters.

P13-4 Four-stroke cycle with a displacement of 0.25 liters.

P13-5 Two-stroke cycle with a displacement of 0.50 liters.

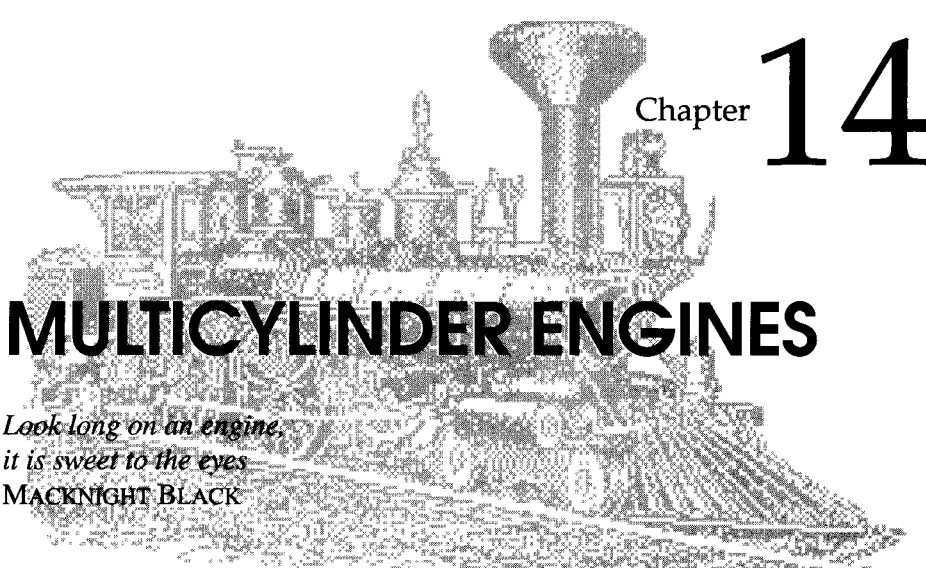
P13-6 Four-stroke cycle with a displacement of 0.50 liters.

P13-7 Two-stroke cycle with a displacement of 0.75 liters.

P13-8 Four-stroke cycle with a displacement of 0.75 liters.

† These problems are suited to solution using Mathcad, Matlab, or TKSolver equation solver programs.

‡ These problems are suited to solution using program ENGINE which is on the attached CD-ROM.



Chapter

14

MULTICYLINDER ENGINES

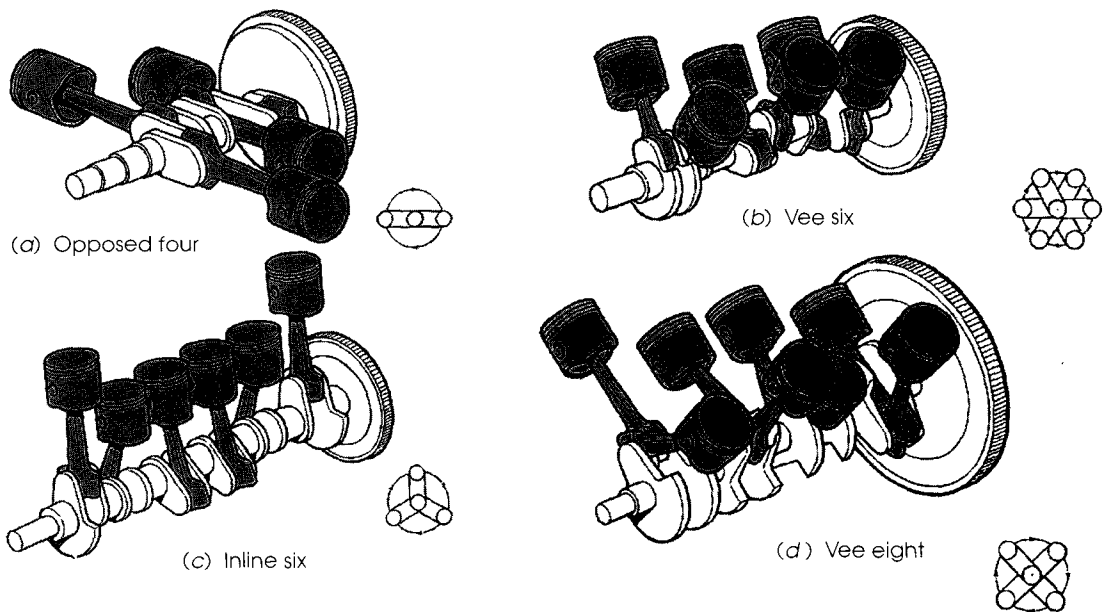
*Look long on an engine,
it is sweet to the eyes*

MACKNIGHT BLACK

14.0 INTRODUCTION

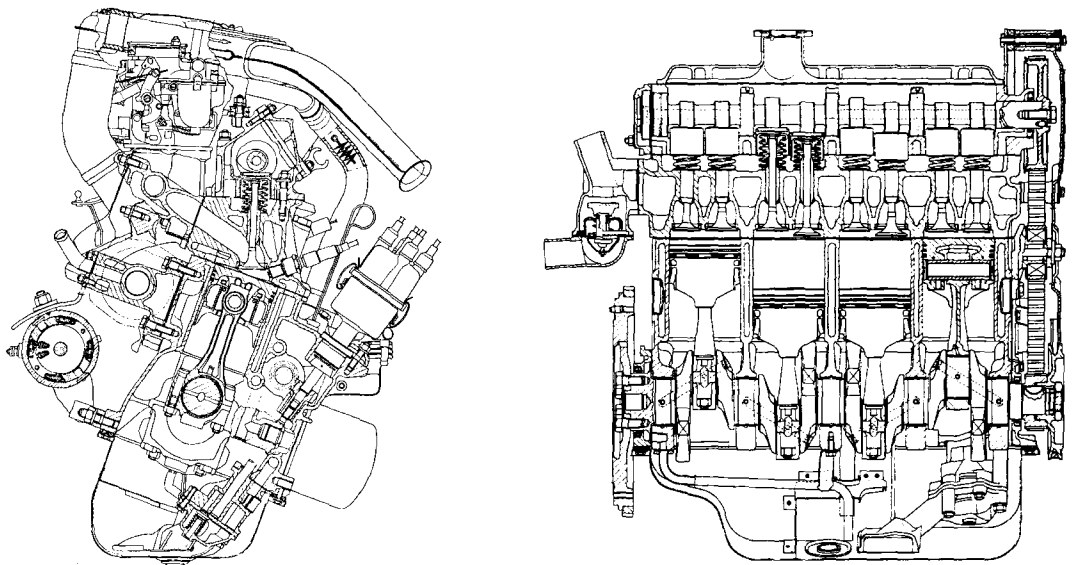
The previous chapter discussed the design of the slider-crank mechanism as used in the single-cylinder internal combustion engine and piston pumps. We will now extend the design to multicylinder configurations. Some of the problems with shaking forces and torques can be alleviated by proper combination of multiple slider-crank linkages on a common crankshaft. Program ENGINE, included with this text, will calculate the equations derived in this chapter and allow the student to exercise many variations of an engine design in a short time. Some examples are provided as disk files to be read into the program. These are noted in the text. The student is encouraged to investigate these examples with program ENGINE in order to develop an understanding of and insight to the subtleties of this topic. A user manual for program ENGINE is provided in Appendix A which can be read or referred to out of sequence, with no loss in continuity, in order to gain familiarity with the program's operation.

As with the single-cylinder case, we will not address the thermodynamic aspects of the internal combustion engine beyond the definition of the combustion forces necessary to drive the device presented in the previous chapter. We will concentrate on the engine's kinematic and mechanical dynamics aspects. It is not our intention to make an "engine designer" of the student so much as to apply dynamic principles to a realistic design problem of general interest and also to convey the complexity and fascination involved in the design of a more complicated dynamic device than the single-cylinder engine.

**FIGURE 14-1**

Various multicylinder engine configurations

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**FIGURE 14-2**

Cutaway views of a four-stroke, four-cylinder inline engine. *Courtesy of FIAT Corporation, Italy*

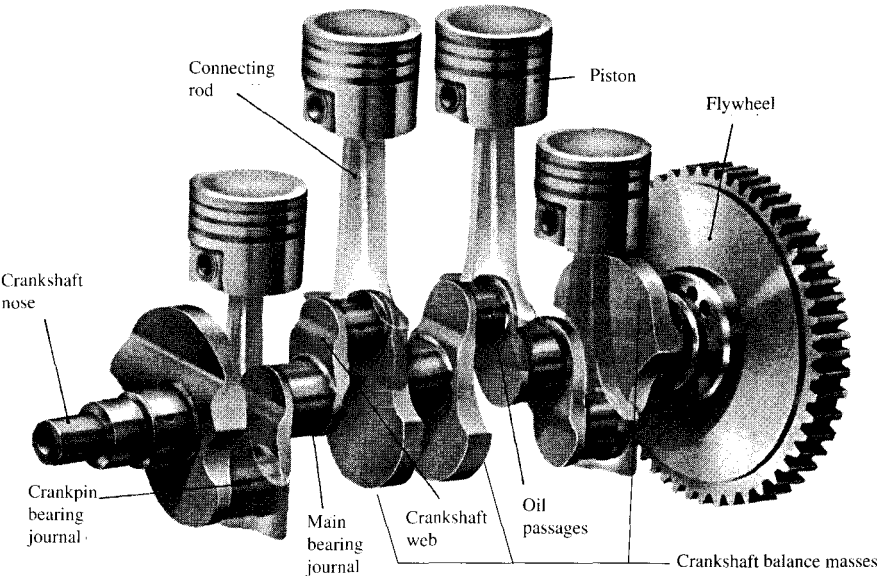


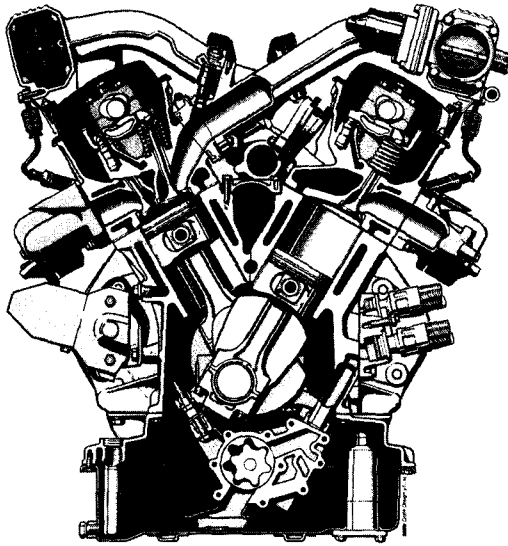
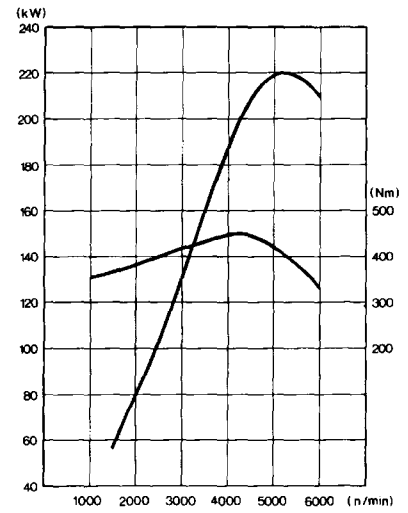
FIGURE 14-3
Crankshaft from an inline four-cylinder engine with pistons, connecting rods, and flywheel
Illustration copyright Eaglemoss Publications/Car Care Magazine. Reprinted with permission.

14.1 MULTICYLINDER ENGINE DESIGNS

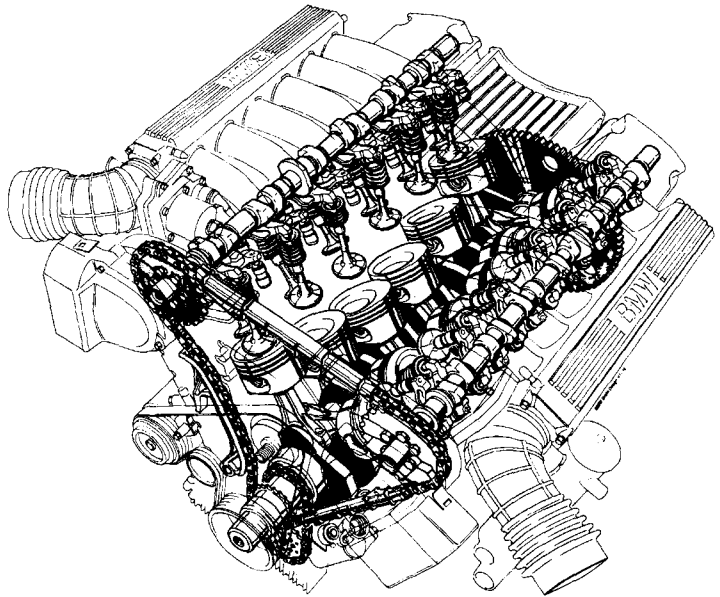
Multicylinder engines are designed in a wide variety of configurations from the simple inline arrangement to vee, opposed, and radial arrangements some of which are shown in Figure 14-1. These arrangements may use any of the stroke cycles discussed in Chapter 13, Clerk, Otto, or Diesel.

INLINE ENGINES The most common and simplest arrangement is an inline engine with its cylinders all in a common plane as shown in Figure 14-2. Two-,* three,* four, five, and six-cylinder **inline engines** are in common use. Each cylinder will have its individual slider-crank mechanism consisting of a crank, conrad, and piston. The cranks are formed together in a common **crankshaft** as shown in Figure 14-3. Each cylinder's crank on the crankshaft is referred to as a **crank throw**. These crank throws will be arranged with some **phase angle** relationship one to the other, in order to stagger the motions of the pistons in time. It should be apparent from the discussion of shaking forces and balancing in the previous chapter that we would like to have pistons moving in opposite directions to one another at the same time in order to cancel the reciprocating inertial forces. The optimum phase angle relationships between the crank throws will differ depending on the number of cylinders and the stroke cycle of the engine. There will usually be one (or a small number at) viable crank throw arrangements for a given engine configuration to accomplish this goal. The engine in Figure 14-2 is a four-stroke cycle, four-cylinder, inline engine with its crank throws at 0, 180, 180, and 0° phase angles which we will soon see are optimum for this engine. Figure 14-3 shows the crankshaft, connecting rods and pistons for the same design of engine as in Figure 14-2.

* Mainly in motorcycles and boats.

**Engine power and torque curves****FIGURE 14-4**

Cross section of a BMW 5-liter V-12 engine and its power and torque curves. *Courtesy of BMW of North America Inc.*

**FIGURE 14-5**

Cutaway view of a 5-liter BMW V-12 engine. *Courtesy of BMW of North America Inc.*

VEE ENGINES in two-, * four-, * six-, eight-, ten-, † and twelve-cylinder+ versions are produced, with vee six and vee eight being the most common configurations. Figure 14-4 shows a cross section and Figure 14-5 a cutaway of a 60° vee -twelve engine. Vee engines can be thought of as *two inline engines grafted together onto a common crankshaft*. The two "inline" portions, or banks, are arranged with some vee angle between them. Figure 14-1d shows a vee-eight engine. Its crank throws are at 0,90,270, and 180° respectively. A vee eight's vee angle is 90°. The geometric arrangements of the crankshaft (phase angles) and cylinders (vee angle) have a significant effect on the dynamic condition of the engine. We will soon explore these relationships in detail.

OPPOSED ENGINES are essentially vee engines with a vee angle of 180°. The pistons in each bank are on opposite sides of the crankshaft as shown in Figure 14-6. This arrangement promotes cancellation of inertial forces and is popular in aircraft engines. § It has also been used in some automotive applications. ||

RADIAL ENGINES have their cylinders arranged radially around the crankshaft in nearly a common plane. These were common on World War II vintage aircraft as they allowed large displacements, and thus high power, in a compact form whose shape was well suited to that of an airplane. Typically air cooled, the cylinder arrangement allowed good exposure of all cylinders to the airstream. Large versions had multiple rows of radial cylinders, rotationally staggered to allow cooling air to reach the back rows. The gas turbine jet engine has rendered these radial aircraft engines obsolete.

ROTARY ENGINES were an interesting variant on the aircraft radial engine. Similar in appearance and cylinder arrangement to the radial engine, the anomaly was that the crankshaft was the stationary ground plane. The propeller was attached to the crankcase (block) which rotated around the crankshaft! It is a kinematic inversion of the radial engine. At least it didn't need a flywheel.

Many other configurations of engines have been tried over the century of development of this ubiquitous device. The bibliography at the end of this chapter contains several references which describe other engine designs, the usual, unusual, and exotic. We will begin our detailed exploration of multicylinder engine design with the simplest configuration, the inline engine, and then progress to the vee and opposed versions.

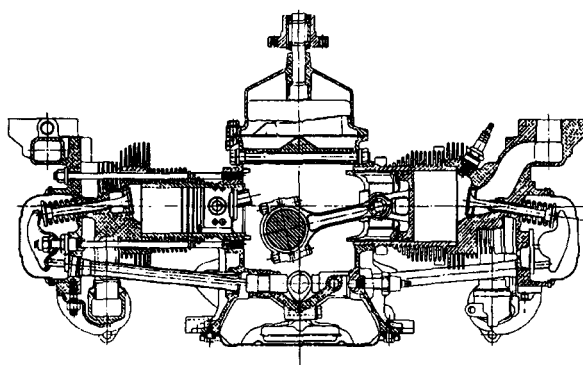


FIGURE 14-6

Chevrolet Corvair horizontally opposed six-cylinder engine
 Courtesy of Chevrolet Division, General Motors Corp.

* Mainly in motorcycles, and boats.

† Honda, Chrysler.

‡ BMW, Jaguar, Mercedes.

§ Continental six-cylinder aircraft engine.

|| Original VW "Beetle" four cylinder, Subaru four, Honda motorcycle six, Ferrari twelve, Porsche six, the ill-fated Corvair six, and the short-lived Tucker (Continental) six, among others.

14.2 THE CRANK PHASE DIAGRAM

Fundamental to the design of any multicylinder engine (or piston pump) is the arrangement of crank throws on the crankshaft. We will use the four-cylinder inline engine as an example. Many choices are possible for the crank phase angles in the four-cylinder engine. We will start, for example, with the one that seems most obvious from a commonsense standpoint. There are 360° in any crankshaft. We have four cylinders, so an arrangement of 0° , 90° , 180° , and 270° seems appropriate. The **delta phase angle** between throws is then 90° . In general, for maximum cancellation of inertia forces, which have a period of one revolution, the optimum delta phase angle will be:

$$\Delta\phi_{\text{inertia}} = \frac{360^\circ}{n} \quad (14.1)$$

where n is the number of cylinders.

We must establish some convention for the measurement of these phase angles which will be:

- 1 The first (front) cylinder will be number 1 and its phase angle will always be zero. It is the reference cylinder for all others.
- 2 The phase angles of all other cylinders will be measured with respect to the crank throw for cylinder 1.
- 3 Phase angles are measured internal to the crankshaft, that is, with respect to a rotating coordinate system embedded in the first crank throw.
- 4 Cylinders will be numbered consecutively from front to back of the engine.

The phase angles are defined in a **crank phase diagram** as shown in Figure 14-7 for a four-cylinder, inline engine. Figure 14-7a shows the crankshaft with the throws numbered clockwise around the axis. The shaft is rotating counterclockwise. The pistons are oscillating horizontally in this diagram, along the x axis. Cylinder 1 is shown with its piston at top dead center (TDC). Taking that position as the starting point for the abscissas (thus time zero) in Figure 14-7b, we plot the velocity of each piston for two revolutions of the crank (to accommodate one complete four-stroke cycle). Piston 2 arrives at TDC 90° after piston 1 has left. Thus we say that cylinder 2 lags cylinder 1 by 90 degrees. By convention a *lagging event is defined as having a negative phase angle*, shown by the clockwise numbering of the crank throws. The velocity plots clearly show that each cylinder arrives at TDC (zero velocity) 90° later than the one before it. Negative velocity on the plots in Figure 14-7b indicates piston motion to the left (down stroke) in Figure 14-7a; positive velocity indicates motion to the right (up stroke).

For the discussion in this chapter we will assume counterclockwise rotation of all crankshafts, and all phase angles will thus be negative. We will, however, omit the negative signs on the listings of phase angles with the understanding that they follow this convention.

Figure 14-7 shows the timing of events in the cycle and is a necessary and useful aid in defining our crankshaft design. However, it is not necessary to go to the trouble of drawing the correct sinusoidal shapes of the velocity plots to obtain the needed information. All that is needed is a schematic indication of the relative positions within the cy-